HOG-ShipCLSNet: A Novel Deep Learning Network With HOG Feature Fusion for SAR Ship Classification

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***Abstract*— Ship classification in synthetic aperture radar (SAR) images is a fundamental and significant step in ocean surveillance. Recently, with the rise of deep learning (DL), modern abstract features from convolutional neural networks (CNNs) have hugely improved SAR ship classification accuracy. However, most existing CNN-based SAR ship classifiers overly rely on abstract features, but uncritically abandon traditional mature hand-crafted features, which may incur some challenges for further improving accuracy. Hence, this article proposes a novel DL network with histogram of oriented gradient (HOG) feature fusion (HOG-ShipCLSNet) for preferable SAR ship classification. In HOG-ShipCLSNet, four mechanisms are proposed to ensure superior classification accuracy, that is, 1) a multiscale classification mechanism (MS-CLS-Mechanism); 2) a global self-attention mechanism (GS-ATT-Mechanism); 3) a fully connected balance mechanism (FC-BAL-Mechanism); and 4) an HOG feature fusion mechanism (HOG-FF-Mechanism). We perform sufficient ablation studies to confirm the effectiveness of these four mechanisms. Finally, our experimental results on two open SAR ship datasets (OpenSARShip and FUSAR-Ship) jointly reveal that HOG-ShipCLSNet dramatically outperforms both modern CNN-based methods and traditional hand-crafted feature methods.**

***Index Terms*— Convolutional neural networks (CNNs), deep learning (DL), feature fusion, histogram of oriented gradient (HOG), ship classification, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) is an essential remote sensing observation tool whose all-day and all-weather working capacity allows it to play a considerable role in maritime domain awareness (MDA) [1]–[8] and ocean surveillance [9]–[16]. As typical participants in marine activities, ships at sea have a great influence on the security, economy, and environment of MDA [8], [17], [18], so SAR ship surveillance is of great value. On the military side, it is conducive to maritime strategic deployments and maritime war early warning [37]. On the civil side, it is also conducive to salvage at sea [74], [75], marine traffic monitoring [76], [77], marine fishery management [2], [8], marine environmental protection [78], [79], and so on. Thus, in recent years, SAR ship surveillance (including ship detection [9]–[14], [20]–[30] and ship classification [33]–[71]) has received much attention from an increasing number of scholars.

Notably, as a fundamental and significant step in ocean surveillance, SAR ship classification can effectively discriminate specific categories of ships and achieve more comprehensive information in marine activities, which is helpful for all-round and effective formulations of marine decision-making.

Since the United States launched the first SAR satellite Seasat, SAR ship classification has been of concern to many scholars [33]–[71]; thus far, various methods have been proposed, including: 1) traditional hand-crafted feature methods and 2) modern convolutional neural network (CNN)-based ones. For the former (i.e., the traditional methods), one always needs to design some hand-crafted features to characterize different ship categories using some mature and explicable mathematical theories. This was a standard solution in earlier years. Nowadays, however, SAR images can be obtained from an increasing number of satellites under various ocean

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Fig. 1. Four mechanisms to ensure the superior classification accuracy of HOG-ShipCLSNet. Note: 1) MS-CLS-Mechanism: multiscale classification mechanism; 2) GS-ATT-Mechanism: global self-attention mechanism; 3) FC-BAL-Mechanism: fully connected balance mechanism; and 4) HOG-FF-Mechanism: HOG feature fusion mechanism.

conditions, various ship resolutions, and so on; these traditional methods are facing great challenges in improving model migration and generalization ability. Meanwhile, repetitive manual model modification is also time-consuming and laborious, with low efficiency. For the latter (i.e., the modern CNN-based methods), one usually does not focus excessively on feature design, because this process is often replaced concisely and efficiently by elegant CNNs. This actually benefits from the record-breaking advantages of artificial intelligence (AI) and deep learning (DL) [80], that is, deep CNNs can enable computational models with multiple processing layers to learn data representations with multiple-level abstractions [80], which can ensure classification accuracy and achieve huge improvements.

Nowadays, there is no doubt at all that abstract features automatically extracted via modern deep CNNs offer better representation ability than the traditional hand-crafted features. However, should we uncritically abandon traditional mature hand-crafted features to rely entirely on abstract ones? This is a meaningful question worth pondering. Regrettably, most existing CNN-based SAR ship classifiers do not offer such profound thinking. Consequently, they may poorly fall into the single cycle of network structure modifications, training trick optimizations, loss function improvements, and so on, potentially creating great challenges for further performance improvements.

Hence, to solve this problem, a novel DL network with the histogram of oriented gradient (HOG) feature fusion (HOG-ShipCLSNet) is proposed for preferable SAR ship classification. In HOG-ShipCLSNet, four mechanisms are proposed to ensure superior classification accuracy [see Fig. 1]. These are: 1) multiscale classification mechanism (MS-CLS-Mechanism); 2) global self-attention mechanism (GS-ATT-Mechanism); 3) fully connected balance mechanism (FC-BAL-Mechanism); and 4) hog feature fusion mechanism (HOG-FF-Mechanism). The first three focus on network structure to improve accuracy, and the last focuses on the fusion between traditional hand-crafted features and modern abstract features to further improve accuracy. To confirm the effectiveness of these four mechanisms, we perform sufficient ablation studies. The experimental results on two open SAR ship datasets (i.e., Open-SARShip [70] and FUSAR-Ship [71]) jointly reveal that HOG-ShipCLSNet achieves a state-of-the-art SAR ship classification accuracy, compared with both modern CNN-based methods and traditional hand-crafted feature methods.

The main contributions of this article are as follows.

1) A novel DL network called HOG-ShipCLSNet is proposed for preferable SAR ship classification.
2) In HOG-ShipCLSNet, four mechanisms are proposed to ensure its superior classification accuracy.
3) In HOG-ShipCLSNet, traditional hand-crafted HOG features and modern abstract CNN features are combined to improve classification accuracy.
4) HOG-ShipCLSNet achieves a state-of-the-art classification accuracy, compared with both modern CNN-based methods and traditional hand-crafted feature methods.

The rest of this article is arranged as follows. Section II reviews the related work. Section III introduces HOG-ShipCLSNet. Section IV introduces the experiments. The results are shown in Section V, and ablation studies are presented in Section VI. Finally, Section VII provides the conclusion.

II. Related Work

In this section, we will review some related works, of which there are two main types: 1) traditional hand-crafted feature methods and 2) modern CNN-based methods.

A. Traditional Hand-Crafted Feature Methods

The most prominent characteristic of the traditional hand-crafted feature methods is manual-feature design, which is often accompanied by some explicable theories, that is, geometric features, texture features, moment features, scattering statistics features, scale-invariant features [73], and HOG features [69].

In earlier years, Musman *et al.* [31], Pastina and Spina [32], and Chao and David [33] made some initial attempts at ship classification in inverse SAR (ISAR) images (e.g., designing a series of distortion-invariant filters to identify ship categories); however, their models often depended heavily on SAR image quality. Afterward, Margarit *et al.* [34], Margarit and Mallorqui [35], and Margarit and Tabasco [36] also conducted some milestone simulation work for SAR ship classification, for example, modeling the geometric features of ship bows, ship middle sections, and stern sections based on their structural information, which inspired a series of follow-up works [37]–[40].

In 2012, Chaturvedi *et al.* [37] drew support from prior information from the Automatic Identification System (AIS) prior to implement automatic "friend" or "foe" ship target recognition from TerraSAR-X images; however, they did not focus on the design of ship features. Later, Zhao *et al.* [38] also concretely classified ships in TerraSAR-X images into three categories, that is, cargo, oil tanker, and container ship. In their model, the analytic hierarchy process (AHP) was applied to select the most appropriate features from the feature

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set (i.e., geometrical structure features, transformation features, or localized invariability features, etc.); then, a K-nearest neighbor (KNN) classifier was used for the final decision.

In 2013, Zhang *et al.* [39] performed a detailed scattering component analysis of merchant ships including bulk carriers, oil tankers, and container ships from COSMO-SkyMed SAR images for their fine classification with an overall accuracy of over 80%. However, their proposed designative ship features have certain drawbacks in the migration applications of other satellites. In addition, Xing *et al.* [40] developed a ship classifier using TerraSAR-X images on the basis of sparse representations in feature space, where the geometric and the local radar cross section (LRCS) density features were used to construct a feature dictionary and sparse representation classification (SRC) was applied to recognize ships. However, they did not consider more complex sea clutter conditions, which may have led to a weak migration and generalization capacity.

In 2014, Wang *et al.* [41] also designed a hierarchical ship classifier for COSMO-SkyMed SAR images, where the geometric and backscattering characteristics of various ship types were analyzed and used; however, they merely collected 41 ship samples that would pose a decline in the model robustness. Later, inspired by the above hierarchical classification concept, Lang *et al.* [42] also designed a hierarchical ship classifier from RadarSAT-2 and TerraSAR-X SAR images, which first extracted dense scale-invariant feature transform (SIFT) [73] as local features and then adopted the k-means clustering algorithm to generate a feature dictionary; finally, a support vector machine (SVM) was used to identify ship categories.

In 2015, Touzi and Vachon [43] performed some AIS ship identification work on the basis of RadarSAT-2 SAR images that provided some tug and ferry transportation information, but they did not design ship classifiers, and ship category information was only from based on the AIS data. Wu *et al.* [44] focused on ship classification from high-resolution SAR images, that is, distinguishing bulk carriers, container ships, and oil tankers. They first designed one three-element feature vector that was composed of kernel density mean features, three-structural features, and a backscattering coefficient estimation mean feature, and then, this feature vector was inputted into one SVM classifier for the final ship classification. However, their model depended on ship resolutions, so it has limited application scenarios.

In 2016, inspired by Zhang *et al.* [39], Jiang *et al.* [45] also divided a ship into bow, middle, and stern to extract their superstructure-scattering features. Then, one SVM was also adopted to perform the final classification. In addition, Lang *et al.* [46] made an extensive study on how to simultaneously select more discriminative features and a more appropriate classifier to improve classification performance. According to their method, the classifier selection strategy was integrated into a wrapper feature selection framework; afterward, a sequential forward floating searching algorithm was used to search for an optimal triplet of a feature–scaling–classifier. Hence, the SAR ship classification accuracy was improved, but the long optimal algorithm search time is bound to reduce efficiency. Later, Leng *et al.* [47] also proposed a 2-D comb feature on the basis of the LRCS statistic to distinguish different types of ships in high-resolution TerraSAR-X SAR images; however, the effectiveness of their method was maintained by having a sufficiently high resolution; otherwise, features would not be represented correctly.

In 2017, inspired by Lang *et al.* [46], Wu *et al.* [48] also studied some joint optimization work on feature selection and classifier design. They proposed a kernel extreme learning machine with a binary dragonfly algorithm (BDA-KELM) to conduct automatic feature selection and search for optimal parameter sets for better ship classification. Similarly, however, the long optimal algorithm search time would also reduce the efficiency. To solve the problem of moderate-resolution SAR ship classification, a series of simpler and efficient naive geometric features were proposed by Lang *et al.* [49], then multiple kernel learning was adopted to learn their combination weights, and finally, one SVM served as a final decision-maker. Zhu *et al.* [50] also combined both topology and intensity of scattering points to enhance SAR ship classification accuracy, but their designed features possibly become unstable for another some unknown ships outside of their datasets.

In 2018, to address the bottleneck of an inadequate amount of labeled data for classifier training, Lang *et al.* [51] drew support from the AIS data to improve the classification accuracy through knowledge transfer. Combined with their previous work in [49], their classifier achieved bigger improvements under the condition of only a few training samples. Particularly, inspired by the SAR-HOG [94] features in SAR automatic target recognition (ATR) community, Lin *et al.* [52] proposed a task-driven dictionary learning (TDDL) framework with the manifold learning SAR-HOG feature (MSHOG) to improve the classification performance. They confirmed that HOG features have better representation ability ($\sim$5% accuracy improvement) than most features, for example, 2-D comb features (2DC) from Leng *et al.* [47], superstructure-scattering features (SS) from Jiang *et al.* [45], and local RCS density features associated with geometry (LRCSG) from Xing *et al.* [40]. Therefore, inspired by their work [52], better HOG features of ships will be fused into our deep network ShipCLSNet to form a new HOG-ShipCLSNet to further enhance CNN-based models' classification performance.

In 2019, Snapir *et al.* [53] reported the research results on the identification of fishing and non-fishing ships in Sentinel-1 SAR images to conduct marine fisheries management. They inputted ship length, longitude, and latitude from the AIS data into a random forest (RF) classifier to make the final decision, which achieved fairish overall classification accuracy. In addition, inspired by Lang *et al.* [51], Xu *et al.* [54] also proposed one discriminative adaptation regularization framework-based transfer learning (D-ARTL) technique based on the unsupervised domain adaptation concept using AIS information to address the problem of insufficient labeled training data. However, there was still a shortcoming of distribution importance imbalance for different classes in their

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method [54], which may decline the benefits of transfer learning.

In 2020, Xu and Lang [55], [56] further optimized their previous work in [54]. For instance, the distance metric learning (DML) algorithm was adopted in [55] to enhance the discriminative ability of feature representations by preserving supervised ship information, which can increase the interclass separability and intraclass compactness to achieve better fine-grained ship classification performance. In [56], they also transferred geometric features based on the metric learning idea to further improve the SAR ship classification performance.

To summarize, the above traditional hand-crafted feature methods have achieved satisfactory results on certain occasions, but they always fall into the dilemma of time-consuming and labor-consuming manual feature design, complex mathematical theories, and weak migration capacity. Fortunately, modern CNN-based methods can smoothly solve the above defects.

B. Modern CNN-Based Methods

The most remarkable characteristic of modern CNN-based methods is automatic feature extraction without any human participation. In general, features extracted by deep CNNs are often abstract, so they are difficult for humans to understand, but they are often regarded as representations of data with multiple levels of abstraction [80]. These multiple levels of abstraction are probably an intrinsic logic of nature, so they can greatly improve the classification performance [80].

Since AlexNet [81] achieved victory in 2012 ImageNet image classification competition [82], the CNN-based classification models have almost completely dominated the field of DL image classification, whose accuracies have far surpassed those of the traditional methods. Thus, in recent years, an increasing number of scholars [57]–[68] in the SAR ship classification community have started to pay more attention to the design CNN-based SAR ship classifiers.

In 2018, Huang *et al.* [70] released OpenSARShip, a large-scale SAR ship classification dataset, dedicated to Sentinel-1 ship interpretation, which provided an important data source for SAR ship recognition using DL. This dataset will be used in this article to confirm the effectiveness of our proposed HOG-ShipCLSNet. Using this dataset, increased research results have started to emerge. For instance, Li *et al.* [57] proposed a dense residual network (DRNet) combining upsampling data augmentation and ratio batching to address the learning imbalance using the above class-unbalanced OpenSARShip dataset; however, their DRNet model was relatively heavy, decreasing training and test efficiency. Bentes *et al.* [58] also proposed a multiple-input resolution CNN-based model to enhance the feature representation ability. In their work, four types of CNN classifiers are used to discriminate cargo, tanker, windmill, platform, and harbor structures. Finally, their experimental results revealed that the CNN-based models greatly outperformed the traditional classifiers (e.g., SVM). Afterward, inspired by the multiple-input resolution idea of Bentes *et al.* [58], Ma *et al.* [59] also designed a single-shot multibox detector with a multiresolution input (MR-SSD) to achieve both ship detection and follow-up specific class discrimination using Gaofen-3 SAR images. They also found that the CNN-based models greatly outperformed the traditional classifiers, for example, SVM and KNN. In addition, Lu and Li [60] established a CNN with data augmentation to improve SAR ship classification on small training dataset; moreover, to avoid overfitting, transfer learning was also adopted. Almost simultaneously, Wang *et al.* [61] conducted some small sample SAR ship classification research based on transfer learning. In their work, classical very deep convolutional networks called "VGGs" [83] with ImageNet pre-training weights [84] were applied to extract the ships' abstract features. However, their long pretraining time is inefficient; moreover, their method is likely to lead to learning bias between different data [29].

In 2019, Shao *et al.* [62] also conducted some work on the class-imbalanced OpenSARShip dataset. They proposed a balanced batch-based sampling method to avoid learning imbalance during training, by which the classification accuracy of the ship class with fewer training samples was improved. Later, Laurentiis *et al.* [63] also applied capsule neural networks (CapsNets) to improving the use of spatial information, in particular of spatial dependence between features. Compared with their raw CNN, a better classification performance was achieved. In addition, Dechesne *et al.* [64] investigated a deep neural network for joint classification and characterization of ships from Sentinel-1 SAR images. Their model was able to distinguish tanker, cargo, fishing, passenger, and tug vessels, but it had the disadvantage of poor generalization because they merely studied the characteristics of ships in space-limited ocean areas, that is, designated ports. Dong *et al.* [65] designed a deep residual network for fine-grained ship classification using Gaofen-3 SAR images; in their reports, they achieved a 99% classification accuracy of cargo, container ships, and oil tankers, but their dataset was too simple to be widely used for migration on other types of SAR images, that is, lack of broad representation.

In 2020, Huang *et al.* [66] comprehensively studied what, where, and how to transfer in SAR ATR. In their work, they improved the SAR ship classification accuracy on OpenSARShip by fine-tuning the deep networks on other datasets, for example, moving and stationary target acquisition and recognition (MSTAR) [85], OpenSARUrban [86], and ImageNet [82]. Afterward, they also proposed a new Deep SAR-Net [67] that considers complex-valued SAR images to learn the spatial texture information and backscattering patterns of ships. Their Deep SAR-Net also achieved knowledge transfer from optical images to SAR images for identifying bulk carriers, container ships, and tankers. Recently, He *et al.* [68] proposed a densely connected triplet CNN for ship classification in medium-resolution SAR images; similar to Xu and Lang [55], they used a DML scheme to expand the distance between different classes. Finally, using their three- and five-class ship recognition datasets, their model achieved

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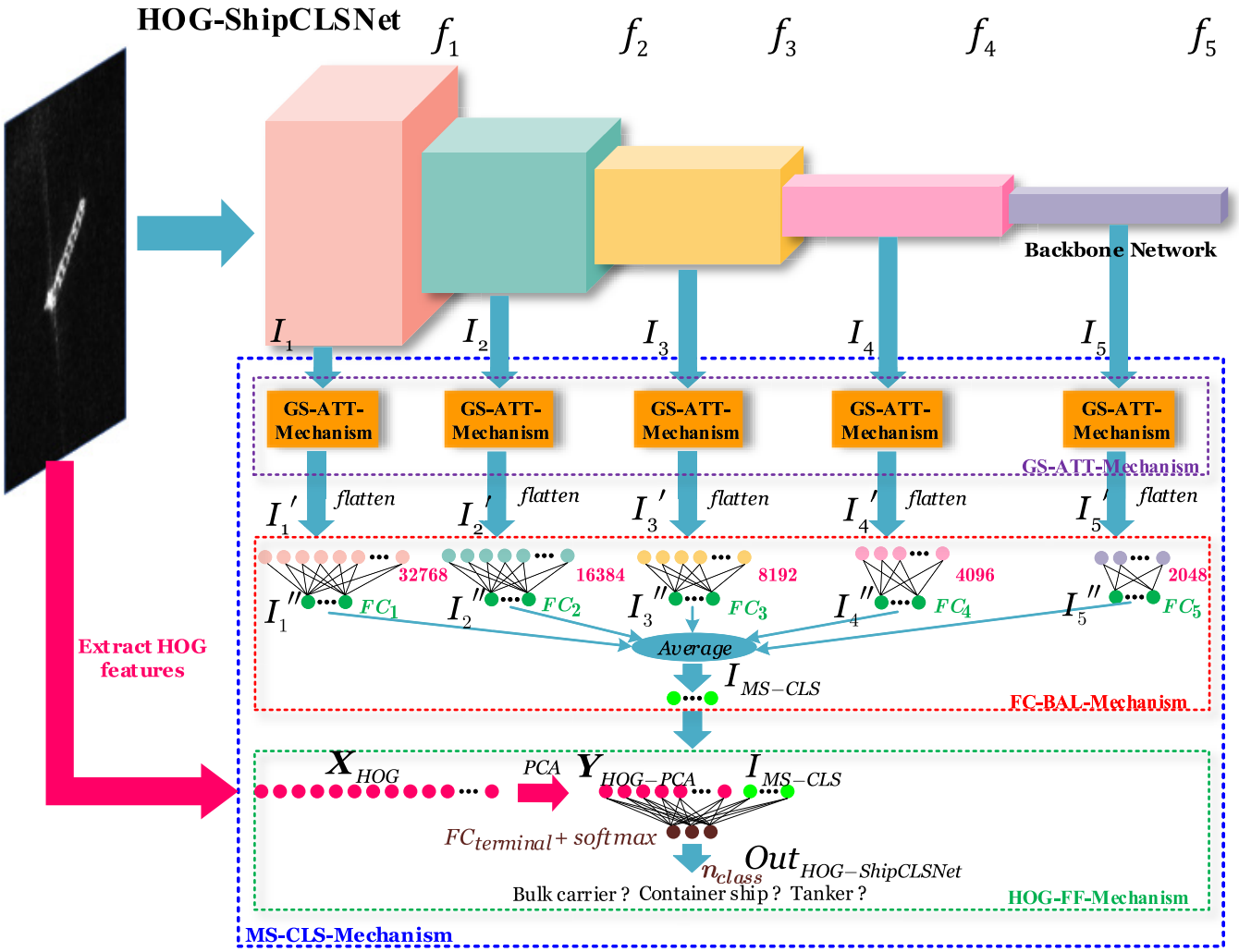


Fig. 2. Overall network architecture of the HOG-ShipCLSNet. MS-CLS-Mechanism: Multiscale classification mechanism; GS-ATT-Mechanism: global self-attention mechanism; FC-BAL-Mechanism: fully connected balance mechanism; and HOG-FF-Mechanism: HOG feature fusion mechanism.

superior performance, but these datasets are still small, which may result in weak robustness and generalization capacity. Particularly, Hou *et al.* [71] also released a high-resolution SAR ship classification dataset named FUSAR-Ship collected from the Gaofen-3 satellite, which will also be conducive to future research by more scholars. In their reports, on the FUSAR-Ship dataset, they also initially established a CNN to achieve fairly reasonable classification results. (This dataset will also be used in this article to verify the effectiveness of our proposed HOG-ShipCLSNet.)

To summarize, DL has become one of the mainstream methods for SAR ship classification. In addition, given the urgent requirements of remote sensing with intelligent processing [72], many researches based on DL are expected to emerge continuously in the future.

Nonetheless, the above CNN-based SAR ship classifiers almost all rely excessively on abstract features while uncritically abandon traditional mature hand-crafted ones. This situation may incur certain challenges in terms of further enhancing the classification performance. Thus, HOG-ShipCLSNet is proposed in this article for preferable SAR ship classification.

III. HOG-ShipCLSNet

HOG-ShipCLSNet comprises two basic components. One is *ShipCLSNet*, and the other is HOG feature fusion. First, *ShipCLSNet* is a brand new network built from scratch. In ShipCLSNet, MS-CLS, GS-ATT, and FC-BAL mechanisms are proposed to guarantee superior performance. These three mechanisms are proposed from the perspective of network structure to improve accuracy. Then, the HOG-FF-Mechanism is designed to formally construct the final HOG-ShipCLSNet to achieve better classification performance. This mechanism is from selected perspective of fusion between the traditional hand-crafted features and modern abstract features to improve accuracy.

Fig. 2 shows the overall network architecture of HOG-ShipCLSNet. Inspired by LeNet-5 [87] and our previous work in [10], we arrange five stages (f_1, f_2, f_3, f_4, and f_5) to extract abstract ship features. These five stages jointly constitute the backbone network of HOG-ShipCLSNet. In addition, these five stages' outputs are denoted by I_1, I_2, I_3, I_4, and I_5 which contain ships' features under five different scales.

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TABLE I
BACKBONE NETWORK ARCHITECTURE OF HOG-SHIPCLSNET

Stage	Layer	Input Shape	Output Shape	Kernel@Stride
f_1	C_1	128×128×1	128×128×8	3×3×8@1
	M_1	128×128×8	64×64×8	@2
f_2	C_2	64×64×8	64×64×16	3×3×16@1
	M_2	64×64×16	32×32×16	@2
f_3	C_3	32×32×16	32×32×32	3×3×32@1
	M_3	32×32×32	16×16×32	@2
f_4	C_4	16×16×32	16×16×64	3×3×64@1
	M_4	16×16×64	8×8×64	@2
f_5	C_5	8×8×64	8×8×128	3×3×128@1
	M_5	8×8×128	4×4×128	@2

Table I shows the backbone network architecture. From Table I, each stage comprises a convolutional layer and a max-pooling layer (C_n and M_n). In addition, the classical rectified linear unit (ReLU) [12] serves as an activation function for all convolutional layers. In Table I, the input or output shape is denoted by $L \times L \times W$ where $L \times L$ denotes the feature map's size and W denotes the channel width (the convolutional kernel number).

First, inspired by VGG [83], we adopt maxpooling layers with a stride of 2 (marked @2) for five times to achieve the size reduction of feature maps with deepening of network. Hence, the strides of all convolutional layers are set to 1 (marked @1), not requiring the size reduction of feature maps.

Then, we also follow the design concept of ResNet [90] to set the convolutional kernels' size to 3×3 to avoid generating superfluous network parameters to decrease the risk of overfitting.

Moreover, with network deepening, to enhance abstract semantic features, we also set the channel width to a bigger and bigger value, reducing the risk of feature loss [10] (i.e., from 8 in C_1, 16 in C_2, 32 in C_3, and 64 in C_4 to 128 in C_5) where width increasing by the power of 2 follows the concept of the existing mature CNN models, for example, VGG [83] and ResNet [90].

Finally, the image input size of HOG-ShipCLSNet is set as 128×128, just as in our previous work [16]; the input channel number of HOG-ShipCLSNet is 1 where 1 refers to the VV or VH channel in OpenSARShip and the FUSAR-Ship dataset. A detailed description of the datasets will be presented in Section IV-A. With the network deepening, the size changes from 128×128 in input, 64×64 in f_1, 32×32 in f_2, 16×16 in f_3, and 8×8 in f_4, to 4×4 in f_5. Furthermore, in HOG-ShipCLSNet, after f_5, we do not further shrink the sizes of the feature maps, because too small-size feature maps run the risk of feature loss; in fact, such practice is also abided by mature CNN models, for example, VGG [83] and ResNet [90].

Briefly, the basic workflow of HOG-ShipCLSNet for SAR ship classification is summarized by the above pseudo codes in Algorithm 1, which can be roughly described by the following seven steps.

Step 1: As shown by lines 1–7 of Algorithm 1, the backbone network conducts ship abstract feature extraction. Its input is the SAR ship image $\boldsymbol{X} \in \mathbb{R}^{128\times128\times1}$ and its outputs are the ship's abstract features I_1, I_2, I_3, I_4, and I_5 coming from the f_1, f_2, f_3, f_4, and f_5 scales. Here, K_{C1}, K_{C2}, K_{C3}, K_{C4}, and K_{C5} denote the convolutional kernels from convolutional layers C_1, C_2, C_3, C_4, and C_5; moreover, $\otimes$ denotes the convolutional operation.

Algorithm 1: HOG-ShipCLSNet Processing of a SAR Ship Image

Input: A SAR ship image $\boldsymbol{X} \in \mathbb{R}^{128\times128\times1}$
begin
1 **do** *ship abstract feature extraction.*
2 $\boldsymbol{I}_1 = \boldsymbol{X} \otimes \boldsymbol{K}_{C1}$
3 $\boldsymbol{I}_2 = \boldsymbol{I}_1 \otimes \boldsymbol{K}_{C2}$
4 $\boldsymbol{I}_3 = \boldsymbol{I}_2 \otimes \boldsymbol{K}_{C3}$
5 $\boldsymbol{I}_4 = \boldsymbol{I}_3 \otimes \boldsymbol{K}_{C4}$
6 $\boldsymbol{I}_5 = \boldsymbol{I}_4 \otimes \boldsymbol{K}_{C5}$
7 **End**
8 **do** *ship HOG feature extraction.*
9 $\boldsymbol{X}_{HOG} = f_{HOG}(\boldsymbol{X})$
10 **End**
11 **do** *PCA*
12 $\boldsymbol{Y}_{HOG-PCA} = f_{PCA}(\boldsymbol{X}_{HOG})$
13 **End**
14 **do** *GS-ATT-Mechanism.*
15 $\boldsymbol{I}'_1 = f_{GS-ATT-Mechanism}(\boldsymbol{I}_1)$
16 $\boldsymbol{I}'_2 = f_{GS-ATT-Mechanism}(\boldsymbol{I}_2)$
17 $\boldsymbol{I}'_3 = f_{GS-ATT-Mechanism}(\boldsymbol{I}_3)$
18 $\boldsymbol{I}'_4 = f_{GS-ATT-Mechanism}(\boldsymbol{I}_4)$
19 $\boldsymbol{I}'_5 = f_{GS-ATT-Mechanism}(\boldsymbol{I}_5)$
20 **End**
21 **do** *FC-BAL-Mechanism.*
22 $\boldsymbol{I}''_1 = FC_1\{flatten(\boldsymbol{I}'_1)\}$
23 $\boldsymbol{I}''_2 = FC_2\{flatten(\boldsymbol{I}'_2)\}$
24 $\boldsymbol{I}''_3 = FC_3\{flatten(\boldsymbol{I}'_3)\}$
25 $\boldsymbol{I}''_4 = FC_4\{flatten(\boldsymbol{I}'_4)\}$
26 $\boldsymbol{I}''_5 = FC_5\{flatten(\boldsymbol{I}'_5)\}$
27 $\boldsymbol{I}_{MS-CLS} = (\boldsymbol{I}''_1 + \boldsymbol{I}''_2 + \boldsymbol{I}''_3 + \boldsymbol{I}''_4 + \boldsymbol{I}''_5)/5$
28 **End**
29 **do** *HOG feature fusion.*
30 $\boldsymbol{I}_{fusion} = \boldsymbol{Y}_{HOG-PCA}$ ©$\boldsymbol{I}_{MS-CLS}$
31 **End**
32 **do** *softmax output.*
33 $\boldsymbol{Out}_{HOG-ShipCLSNet} = softmax\{FC_{terminal}(\boldsymbol{I}_{fusion})\}$
34 **End**
35 **end**
Output: Classification tensor $\boldsymbol{Out}_{HOG-ShipCLSNet} \in \mathbb{R}^{nclass\times1}$

Step 2: As shown by lines 8–10 of Algorithm 1, we conduct the SAR ship HOG feature extraction. Its input is the SAR ship image $\boldsymbol{X} \in \mathbb{R}^{128\times128\times1}$ and its output is the ship's HOG features $\boldsymbol{X}_{\mathrm{HOG}}$. More details will be introduced in Section III-D.

Step 3: As shown by lines 11–13 of Algorithm 1, principal component analysis (PCA) [93] is conducted on the ship's HOG features X_{HOG}. Its input is the ship's HOG features $\boldsymbol{X}_{HOG}$, and its output is the reduced ship's HOG features $Y_{\mathrm{HOG-PCA}}$. More details will be introduced in Section III-D.

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Step 4: As shown by lines 14–20 of Algorithm 1, the GS-ATT-Mechanism is applied to the ship features from the backbone network. Its inputs are the ship's abstract features I_1, I_2, I_3, I_4, and I_5 coming from the f_1, f_2, f_3, f_4, and f_5 scales, and its outputs are the refined ship features I_1', I_2', I_3', I_4', and I_5'. More details will be introduced in Section III-B.

Step 5: As shown by lines 21–28 of Algorithm 1, the FC-BAL-Mechanism is conducted on the refined features from the GS-ATT-Mechanism. Its inputs are the refined ship features I_1', I_2', I_3', I_4', and I_5', and its output is the multiscale synthetic features $I_{\mathrm{MS-CLS}}$. Here, the refined ship features I_1', I_2', I_3', I_4', and I_5' are first flattened into column vectors and then inputted into five fully connected (FC) layers (i.e., FC_1, FC_2, FC_3, FC_4, and FC_5 in Fig. 2) for processing. Finally, the outputs of the five FC layers are the conducted average operation to obtain the final CNN abstract feature vector $I_{\mathrm{MS-CLS}}$. In addition, ReLU [12] serves as activation functions of FC_1, FC_2, FC_3, FC_4, and FC_5. More details will be introduced in Section III-C.

Step 6: As shown by lines 29–31 of Algorithm 1, we perform HOG feature fusion. Its inputs are the reduced ship HOG features $Y_{\mathrm{HOG-PCA}}$ and the multiscale CNN features $I_{\mathrm{MS-CLS}}$, and its output is the fused ship feature I_{fusion}. Here, ©in line 30 denotes the concatenation operation. More details will be introduced in Section III-D.

Step 7: An FC layer ($\mathrm{FC}_{\mathrm{terminal}}$) and a softmax function are used to generate the final output. Its input is the fused ship feature I_{fusion}, and its output is the network's output $\mathrm{Out}_{\mathrm{HOG-ShipCLSNet}} \in \mathbb{R}^{n_{\mathrm{class}}\times 1}$, where n_{class} denotes the number of neurons in the terminal FC layer ($\mathrm{FC}_{\mathrm{terminal}}$), equal to ship category number, that is, $n_{\mathrm{class}} = 3$ on OpenSARship and $n_{\mathrm{class}} = 7$ on FUSAR-Ship. In addition, ReLU [12] serves as an activation function for $\mathrm{FC}_{\mathrm{terminal}}$.

Next, we will introduce the four proposed mechanisms in Section III-A–III-D.

A. Mechanism 1: MS-CLS-Mechanism

In HOG-ShipCLSNet, inspired by the feature pyramid network (FPN) [99] from the object detection community, we propose an MS-CLS-Mechanism to enrich the features used for final classification among the terminal FC layers ($\mathrm{FC}_{\mathrm{terminal}}$), as shown in Fig. 2 (marked by a blue dotted box).

Most existing CNN-based models merely adopt the features of the last stage or scale f_5 as the final classification features (i.e., the backend of the networks.), so they all belong to single-scale classification models. However, the front-end features of networks (from f_1 to f_4) contain more small-ship information. Here, the small ship is measured at the pixel level, instead of the actual physical size, which is universally adopted in the computer vision (CV) community of DL. A further introduction to SAR ship size can be found in our previous work in [14] and that of Zhang *et al.* [100]. If the front-end features are thoughtlessly abandoned, the classification accuracy of small ships will become worse. Especially, when the resolution of SAR ship images is relatively low, the front-ends of the networks can also retain more ship features, decreasing the feature loss risk [10]. Moreover, although the shallow features from the network's front-end can be transmitted to subsequent deeper layers, they are likely to be diluted, which may lead to the weakening of the final feature's spatial representation ability.

From Fig. 2, the whole framework of the HOG-ShipCLSNet is based on the MS-CLS-Mechanism. Namely, in Fig. 2, all stages or all scales (f_1, f_2, f_3, f_4, and f_5) are responsible for final ship classification decision. In other words, different scale features of ships from different network depths are used to achieve multiscale feature fusion to ensure better feature representation ability.

It should be noted that our proposed MS-CLS-Mechanism does not adopt the original practice of FPN (i.e., multioutput terminals.); in contrast, we integrate five inputs (I_1'', I_2'', I_3'', I_4'', and I_5'') into one output with the aim of more stable training (i.e., single-output terminal) [92], described by

$$I_{\mathrm{MS-CLS}} = \frac{1}{5} \cdot \sum_{i=1}^{5} I''_i \tag{1}$$

where FC_i denotes the FC layer of the ith classification scale, I_i' denotes the input of the ith classification scale, flatten $(\cdot)$ denotes reshaping into a long column vector (from$\mathbb{R}^{64\times 64\times 8}$ to $\mathbb{R}^{32\,768\times 1}$ for the f_1 scale, from $\mathbb{R}^{32\times 32\times 16}$ to $\mathbb{R}^{16\,384\times 1}$ for the f_2 scale, from $\mathbb{R}^{16\times 16\times 32}$ to $\mathbb{R}^{8192\times 1}$ for the f_3 scale, from $\mathbb{R}^{8\times 8\times 64}$ to $\mathbb{R}^{4096\times 1}$ for the f_4 scale, and from $\mathbb{R}^{4\times 4\times 128}$ to $\mathbb{R}^{2048\times 1}$ for the f_5 scale), and $I_{\mathrm{MS-CLS}}$ denotes the final multiscale CNN abstract features. In (1), we also adopt the average weighting operation to obtain the single-output terminal, as inspired by our previous work [15] and [92], because the experimental results in Section IV-A confirm that the average weighting type is better than the adaptive weighting one, as it comes from more focused ship feature learning that is not distracted by searching weighted parameters.

Moreover, we consider that the ship classification task always contains only one ship in an SAR image, such that the network should only output one classification result/label. If the network outputs several results: 1) during training, it will lead to training oscillation and 2) during testing, it may also lead to decision failure. For example, for a three-classification task (i.e., classes 0, 1, and 2), if the outputs from five scales are classes 0, 1, and 2, then the network will fall into training oscillation because of decision failure if one uses a voting mechanism (i.e., confusion between classes 0 and 1). In addition, multioutput terminals may also incur the risk of decision-making for those ships with large feature differences at different scales.

More details concerning multi-output and single-output forms will be studied by our experiments in Section VI-A.

B. Mechanism 2: GS-ATT-Mechanism

To make the features from different scales more discriminative, we also propose a GS-ATT-Mechanism to further refine these features from the backbone network further to enhance their global expression ability [88], (i.e., to pay more attention to important spatial global information), as is shown in Fig. 2 (marked in five orange boxes). From Fig. 2, the features from every scale are all refined by our proposed GS-ATT-Mechanism to ensure more adequate global response.

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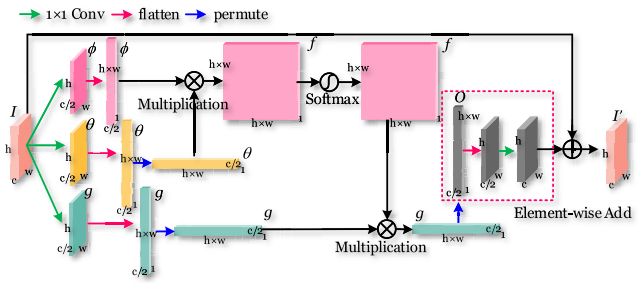


Fig. 3. GS-ATT-Mechanism.

The GS-ATT-Mechanism can be described by

$$O_i = \frac{1}{\mathcal{C}(I)} \cdot \sum_{\forall j} (f(I_i, I_j) \cdot g(I_j)) \tag{2}$$

where I_i denotes the input at the ith location, O_i denotes the output at the ith location, $f(\cdot)$ is a function used to calculate the similarity between I_i and I_j, $g(\cdot)$ is also a function used to characterize the feature representation at the jth location, and $\mathcal{C}(\cdot)$ denotes a normalized coefficient, that is, the input response. Here, the ith location information denotes the response of the current location and the jth location information denotes the global response.

In (2), in general, $g(\cdot)$ can be seen as a linear embedding

$$g(I_j) = W_g I_j \tag{3}$$

where W_g is a weight matrix to be learned, and we use a 1×1 convolutional layer to achieve this weight matrix in training.

Furthermore, similar to Vaswani *et al.* [89], a simple extension of the Gaussian function is used to compute similarity $f(\cdot)$ in an embedding space

$$f(I_i, I_j) = e^{\theta(I_i)^T \phi(I_j)} \tag{4}$$

where $\theta(I_i) = W_\theta I_i$ and $\phi(I_j) = W_\phi I_j$ are two embeddings, W_θ and W_ϕ are the weight matrixes to be learned, and they can both be achieved by another two 1×1 convolutional layers.

As above, the normalized coefficient $\mathcal{C}(\cdot)$ is set by

$$\mathcal{C}(I) = \sum_{\forall j} f(I_i, I_j). \tag{5}$$

Finally, the GS-ATT-Mechanism is instantiated as

$$O_i = \frac{e^{\theta(I_i)^T \phi(I_j)} \cdot W_g I_j}{\sum_{\forall j} e^{\theta(I_i)^T \phi(I_j)}} \tag{6}$$

where $e^{\theta(I_i)^T \phi(I_j)} / \sum_{\forall j} e^{\theta(I_i)^T \phi(I_j)}$ can be achieved through a softmax function/layer.

Fig. 3 shows a graphical description of the above GS-ATT-Mechanism. From Fig. 3, two 1×1 convolutional layers are used to compute ϕ and θ, and then by matrix multiplication $\theta^T \phi$, the similarity f is obtained. One 1×1 convolutional layer is used to characterize the representation of features $g(\cdot)$ (i.e., g). Finally, f through a softmax function/layer with a sigmoid activation is multiplied by g to obtain the self-attention output O.

Particularly, inspired by Wang *et al.* [88], the self-attention output O is also further processed by a 1×1 convolutional layer (marked in a dotted box), whose purpose is to make O match the dimension of the original input I to facilitate follow-up element-wise adding, which is similar to the residual/skip connections in ResNet [90], [91], that is,

$$I' = W_O O + I. \tag{7}$$

Here, W_O is also a weighting matrix to be learned, and another 1×1 convolutional layer can be used to achieve it during training. Hence, finally, the refined features I' combining self-attention information are obtained and will be further processed in subsequent steps.

In essence, the above GS-ATT-Mechanism can directly capture the long-range dependence of each location (i.e., global response) by calculating the interaction between two arbitrary different positions, which is equivalent to constructing a convolutional kernel with the same size as the feature map I, to maintain more useful ship information that makes the feature maps more discriminative. More details can be found in [89].

C. Mechanism 3: FC-BAL-Mechanism

From Fig. 2, the outputs of five stages (f_1, f_2, f_3, f_4, and f_5) are denoted by I_1, I_2, I_3, I_4, and I_5. From Table I, the feature dimension of I_1 is $64 \times 64 \times 8 = 32\,768$ (the output shape of M_1 of f_1 in Table I), that of I_2 is $32 \times 32 \times 16 = 16\,384$ (the output shape of M_2 of f_2 in Table I), that of I_3 is $16 \times 16 \times 32 = 8192$ (the output shape of M_3 of f_3 in Table I), that of I_4 is $8 \times 8 \times 64 = 4096$ (the output shape of M_4 of f_4 in Table I), and that of I_5 is $4 \times 4 \times 128 = 2048$ (the output shape of M_5 of f_5 in Table I).

Thus, one can find that the classification contributions of different scales obviously differ because f_1 has 32 768 features, f_2 has 16 384 features, f_3 has 8192 features, f_4 has 4096 features, and f_5 has 2048 features. Such feature-number imbalance from different scales may cause unstable learning. Hence, to solve this problem, after feature maps are refined by the GS-ATT-Mechanism, we also propose an FC-BAL-Mechanism to further integrate feature maps of different scales I into a unified feature dimension to balance the classification contributions of different scales, as shown in Fig. 2 (marked in the red dotted box). In the FC-BAL-Mechanism, we add an FC layer after each scale, respectively, to achieve feature balance, that is,

$$\begin{aligned}
I_1'' &= \mathrm{FC}_1(I_1') \quad 32\,768 \to N_{\mathrm{FC}} \\
I_2'' &= \mathrm{FC}_2(I_2') \quad 16\,384 \to N_{\mathrm{FC}} \\
I_3'' &= \mathrm{FC}_3(I_3') \quad 8192 \to N_{\mathrm{FC}} \\
I_4'' &= \mathrm{FC}_4(I_4') \quad 4096 \to N_{\mathrm{FC}} \\
I_5'' &= \mathrm{FC}_5(I_5') \quad 2048 \to N_{\mathrm{FC}}
\end{aligned} \tag{8}$$

where the number of neurons of all FC layers (FC_1, FC_2, FC_3, FC_4, and FC_5) from different scales are set to the same value. Hence, features from different scales are unified into the same N_{FC} dimension, meaning that the classification contributions of different scales are equal. Here, N_{FC} is an experimental

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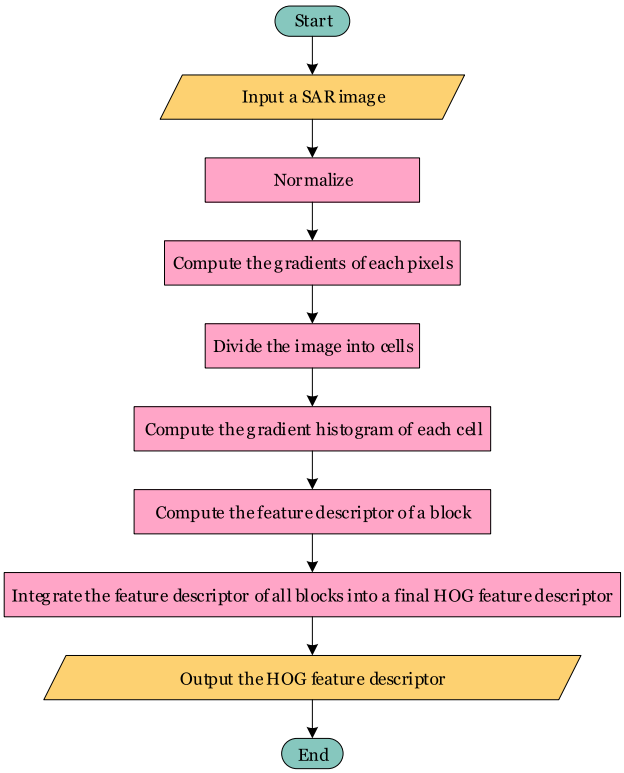


Fig. 4. Process of extracting HOG features.

hyper-parameter in our HOG-ShipCLSNet, which will be experimentally studied in detail in Section VI-C.

D. Mechanism 4: HOG-FF-Mechanism

The HOG-FF-Mechanism is composed of three basic steps, that is: 1) extract HOG features; 2) PCA [93] dimension reduction; and 3) HOG feature fusion.

1) Extract HOG Features: The HOG feature was first proposed by Dalal *et al.* [69] and has been widely used in the pedestrian detection community [101] in early years. Considering that the gradient mainly exists at the edge of the image, the shape and edge information of local targets can be well-described by the gradient [69], which can be used to characterize SAR ship features. Thus far, some scholars [52], [94] from the SAR ATR community have applied HOG feature to perform ship classification, and they have achieved reasonably fair results (although their methods were all traditional hand-crafted feature-based techniques.). Hence, inspired by their work (i.e., using HOG feature to classify ships), we will fuse HOG features into abstract CNN-based features to further improve the classification performance by realizing the mutual complementarity of tradition and modernity. Fig. 4 shows the process of extracting HOG features. Its detailed description is as follows:

Step 1: Input an SAR image.

As in our previous work in [16], each image is reshaped to a 128×128 size by image sampling using the commonly used and classical bilinear interpolation/resampling algorithm [95] to keep the feature dimension the same. In addition, the 128×128 image size is also the same as the input size of the backbone network to ensure the rationality of the follow-up feature fusion.

Step 2: Normalize.

Similar to Dalal *et al.* [69], we adopt the adaptive gamma correction method [96] to normalize the input SAR image into [0, 1] to weaken the interference of speckle noise and reduce the negative impact of SAR image local violent steepness.

Step 3: Compute gradients.

Compute the gradients of each pixel including the amplitude and direction, that is,

$$G(x, y) = \sqrt{G_x^2(x, y) + G_y^2(x, y)} \tag{9}$$

$$\alpha(x, y) = \arctan\left\{\frac{G_x(x, y)}{G_y(x, y)}\right\} \tag{10}$$

where $G_x(x, y)$ denotes the gradient amplitude in the x-direction, and $G_y(x, y)$ denotes that in the y-direction, that is,

$$G_x(x, y) = H(x + 1, y) - H(x - 1, y) \tag{11}$$

$$G_y(x, y) = H(x, y + 1) - H(x, y - 1). \tag{12}$$

Here, $H(i, j)$ denotes the gray value of the ith row and jth column in an image. In (9), $G(x, y)$ denotes the final synthetic amplitude, and in (10), $\alpha(x, y)$ denotes the gradient direction, ranging from 0° to 360° (i.e., from 0° to 180° and the opposite direction).

Step 4: Divide the image into cells.

Inspired by Lin *et al.* [52] and Su *et al.* [97], we divide the image into a number of small cells, each containing 64 pixels (i.e., 8×8), where 8×8 is an experimental optimal value for optimal performance. Consequently, for a 128×128 sized SAR image, 256 cells are generated, where each cell will be analyzed and then used to represent one local feature.

Step 5: Compute the gradient histogram of each cell.

Similar to Lin *et al.* [52], we first divide the gradient direction of each cell into 12 bins—that is, each bin is 30° (i.e., 360°/12)—and then compute the gradient histogram of each cell for each bin. Furthermore, the gradient amplitude needs to be weighted into gradient histogram to maintain the importance of different local regions [69].

Step 6: Compute the feature descriptor of a block.

Similar to Lin *et al.* [52], we first form each set of four cells into a block (i.e., 16×16 pixels for each block), and then normalize the gradient histogram of each cell to weaken the interference of speckle noise and reduce the negative impact of SAR image local violent steepness. Finally, the gradient histograms in each cell among each block are concatenated to construct the block's final feature descriptor.

Step 7: Integrate the feature descriptors of all blocks into the final HOG feature descriptor.

Inspired by Lin *et al.* [52], we take the cell size as the block stride to slide windows in the whole image to form different blocks. Then, the feature descriptors of all blocks are concatenated to obtain the final HOG feature descriptor of an SAR ship image. Given the above parameter configuration, the final HOG feature dimension is 32 448 (see [69] for detailed calculation).

Step 8: Output the HOG feature descriptor.

The final HOG feature descriptor obtained will be integrated with modern CNN-based abstract features to improve the

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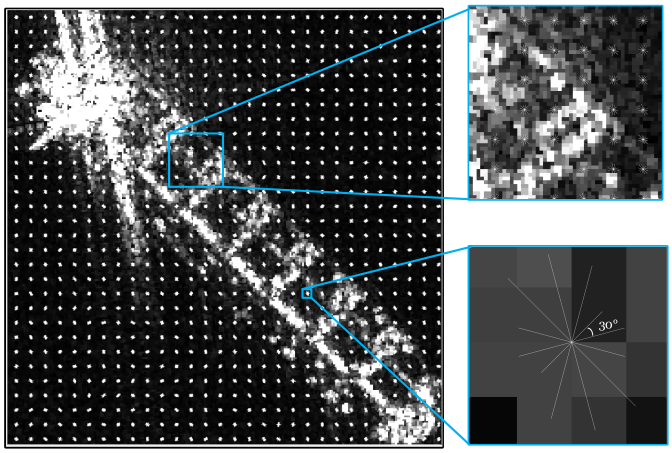


Fig. 5. Visualization of SAR ship HOG features.

classification performance. Fig. 5 shows the HOG feature visualization results of an SAR ship.

More details of the HOG feature extraction process can be found in [52], [69], and [94].

2) PCA Dimension Reduction: Several previous research findings in [52], [69], and [94] indicated that the high-dimension HOG feature descriptor is more likely to bring overfitting; thus, to reduce such overfitting risk, we adopt a commonly used PCA [93] to achieve dimension reduction, that is, from 32 448 dimensions to k, where $k < 32\,448$ is an experimental optimum value that will be experimentally studied in Section VI-D. The PCA flow is shown in Algorithm 2.

Algorithm 2: PCA Dimension Reduction of HOG Features

Input: HOG features $\boldsymbol{X}_{HOG} = (x_1, x_2, \cdots, x_{32448})^T$

Begin

1 Feature centralization: $x_i \leftarrow x_i - \hat{\boldsymbol{X}}_{HOG}$

2 Calculate the covariance matrix $\boldsymbol{M}_{Cov} = \boldsymbol{X}_{HOG}\, \boldsymbol{X}_{HOG}^T$

3 Conduct eigenvalue decomposition on the covariance matrix $\boldsymbol{M}_{Cov}$

4 Select the eigenvector with the largest k eigenvalues $y_1, y_2, \cdots, y_k$

5 **End**

Output: Reduced HOG features $\boldsymbol{Y}_{HOG-PCA} = (y_1, y_2, \cdots, y_k)^T$

Ultimately, the reduced HOG features $Y_{\mathrm{HOG-PCA}}$ will be formally fused into modern CNN-based abstract features to further improve the classification performance. In our experiments, we use Scikit-learn [104] , a common Python-based machine learning library to perform the above PCA operation.

3) HOG Feature Fusion: As shown in Fig. 2, the reduced HOG features $Y_{\mathrm{HOG-PCA}}$ and the CNN-based abstract features $I_{\mathrm{MS-CLS}}$ from (1) are concatenated into one new feature vector, which is then used by the terminal FC layer and a softmax layer for the final classification decision, that is,

$$\mathrm{Out}_{\mathrm{HOG-ShipCLSNet}} = \mathrm{softmax}\{\mathrm{FC}_{\mathrm{terminal}}(\boldsymbol{Y}_{\mathrm{HOG-PCA}} \copyright I_{\mathrm{MS-CLS}})\} \quad (13)$$

TABLE II
SHIP SAMPLE NUMBERS OF EACH CATEGORY IN OPENSARSHIP. # DENOTES THE NUMBER

Dataset	Category	#Training	#Test	#All
OpenSARShip	Bulk carrier	338	328	666
	Container ship	338	808	1146
	Tanker	338	146	484

where $\mathrm{Out}_{\mathrm{HOG-ShipCLSNet}}$ means the output of HOG-ShipCLSNet, ©denotes the concatenation operation, $\mathrm{FC}_{\mathrm{terminal}}$ denotes the last terminal FC layer in Fig. 2, and the softmax layer is defined by $e^{X_i}/\sum_j e^{X_j}$ [98].

IV. EXPERIMENTS

Our experiments are run on a personal computer (PC) with Intel i9-9900K CPU, NVIDIA RTX2080Ti GPU, and 32G memory. Software development is conducted under the Python language environment using an open-sourced machine learning library Pytorch [103]. CUDA10.1 is used to call GPU for training acceleration.

A. Dataset

1) OpenSARShip: The OpenSARShip dataset was released by Huang *et al.* [70] in 2018 with a ship resolution of ∼20 m in range and azimuth. It has been used by many scholars, for example, Wang *et al.* [16], Li *et al.* [57], Shao *et al.* [62], and Huang *et al.* [66], [67]. Therefore, we also use OpenSARShip to confirm the effectiveness of HOG-ShipCLSNet. In our work, OpenSARShip contains three main categories, that is, bulk carrier, container ship, and tanker, which cover approximately 80% of the international shipping market [39], [70].

The OpenSARShip dataset's configuration and some preprocessing mechanisms are the same as in our previous work [16]. Specifically, the single-look complex (SLC) model of Sentinel-1 is used. The ground range multilook detected (GRD) model can be studied in the future. VV and VH polarization SAR images jointly constitute the final dataset used in this article, where the sample number of VV is equal to that of VH. The input dimension of HOG-ShipCLSNet is $128 \times 128 \times 1$ where 1 refers to VV channel or VH channel.

It should be noted that single-channel VV SAR images ($128 \times 128 \times 1$) and single-channel VH SAR images ($128 \times 128 \times 1$) are successively inputted into HOG-ShipCLSNet directly. In other words, they are not combined as the $128 \times 128 \times 2$ dimension. In short, HOG-ShipCLSNet can classify both VV ships and VH ships.

We find that the sample numbers of the three ship categories are imbalanced in OpenSARShip. Therefore, to prevent the adverse effects of class imbalance, we set the number of training samples to be equal for each class, according to the least number of samples in all three categories with the training–testing ratio as 7:3, as in [16]. Moreover, the remaining samples are regarded as testing samples.

Table II shows the ship sample numbers of each category in OpenSARShip. Fig. 6 shows some SAR ship samples in OpenSARShip.

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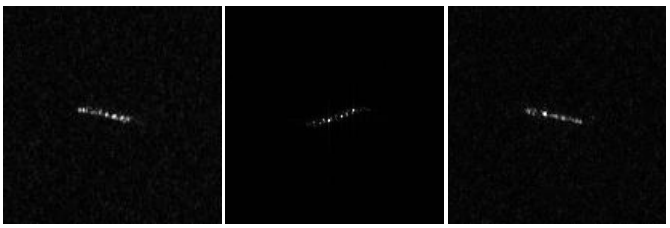


Fig. 6. SAR ship samples in OpenSARShip. (a) Bulk carrier. (b) Container ship. (c) Tanker.

TABLE III
SHIP SAMPLE NUMBERS OF EACH CATEGORY IN FUSAR-SHIP. # DENOTES THE NUMBER

Dataset	Category	#Training	#Test	#All
FUSAR-Ship	Bulk carrier	1150	494	1644
	Container ship	1219	523	1742
	Fishing	1101	473	1574
	Tanker	1215	521	1736
	General cargo	1205	517	1722
	Other cargo	1214	521	1735
	Other	1211	520	1731

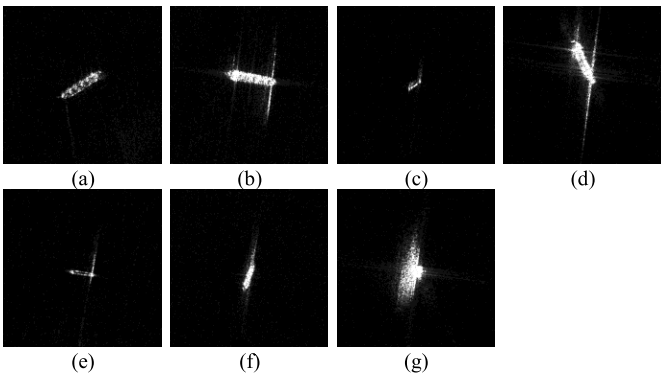


Fig. 7. SAR ship samples in FUSAR-Ship. (a) Bulk carrier. (b) Container ship. (c) Fishing. (d) Tanker. (e) General cargo. (f) Other cargo. (g) Other.

2) FUSAR-Ship: Moreover, to further confirm the effectiveness of HOG-ShipCLSNet, we conduct experiments on the FUSAR-Ship dataset. The dataset was released by Hou *et al.* [71] in 2020 for Gaofen-3 SAR ship interpretation; its ship resolution is ∼1.5 m in range and azimuth.

In this article, FUSAR-Ship mainly contains seven categories, that is, bulk carriers, container ships, fishing, tankers, general cargo ships, other cargo ships, and other. Furthermore, it should be noted that the original FUSAR-Ship dataset contains an extra "false alarm" category. However, this category is not considered in our work because we focus on identifying ship types, rather than discriminating between false alarms and real ships.

Table III shows the ship sample numbers of each category in FUSAR-Ship. Fig. 7 shows some SAR ship samples in FUSAR-Ship.

B. Experimental Details

As in [16], SAR images are resized to 128×128 pixels for input to HOG-ShipCLSNet for abstract feature extraction and traditional HOG feature extraction. To mitigate the negative effects of a small number of samples, the random cropping data augmentation technique is adopted to expand sample diversity. Other data augmentation techniques are also studied in Section V-A.

When running training on the OpenSARShip dataset, the number of neurons of the terminal FC layer ($\mathrm{FC}_{\mathrm{terminal}}$) in HOG-ShipCLSNet is set to 3, which represents the ship categories. When running training on the FUSAR-Ship dataset, the number of neurons of the terminal FC layer in HOG-ShipCLSNet is 7, which represents the case of seven ship categories.

Similar to Wang *et al.* [16], the adaptive moment estimation (Adam) optimizer [104] is used to train HOG-ShipCLSNet for 100 epochs. Considering the limited GPU memory, the training batch size is set to an empirical value of 64. Moreover, we find that the training batch size should not be too small; otherwise, the training-oscillation phenomenon will appear. To mitigate the negative effects of vanishing training gradients, we set the learning rate as a relatively small value of 0.0002 because we find that such a learning rate allows the training to converge more readily. This also comes from the experience of [16].

HOG-ShipCLSNet is trained from scratch without pretraining either in the standard way or via an autoencoder. In addition, as in our previous work [10], we initialize the network parameters using the Glorot uniform distribution [105], also called Xavier uniform initialization.

C. Loss Function

Similar to [16], the cross entropy (CE) loss [16] serves as the loss function of HOG-ShipCLSNet, defined by

$$L_s = -\frac{1}{N_x}\sum_{i=1}^{N_x} y_i' \log(y_i) \tag{14}$$

where y_i denotes the ith sample classification result, y'_i denotes the ith sample ground truth label, and N_x denotes the number of all training samples.

Moreover, HOG feature extraction and PCA processes are both completed before training; this means that $Y_{\mathrm{HOG-PCA}}$ is constant during training, and the parameters in $\mathrm{FC}_{\mathrm{terminal}}$ are trainable. The backpropagation of the loss is through the entire HOG-ShipCLSNet except for HOG feature extraction and PCA processes, and its backpropagation direction is opposite to the blue arrows in Fig. 2. Therefore, after completing these processes, the training is end to end.

D. Evaluation Indices

Accuracy is defined by

$$\text{Accuracy} = \frac{\#\mathrm{TP} + \#\mathrm{TN}}{\#\mathrm{TP} + \#\mathrm{TN} + \#\mathrm{FP} + \#\mathrm{FN}} \tag{15}$$

where # denotes the number. TP denotes the true positives, TN denotes the false positives, FN denotes the false negatives, and TN denotes the true negatives. Briefly, the number of correct ship classifications (N_{correct}) is #TP + #TN and the total number of all ships (N_{all}) is #TP + #TN + #FP + #FN.

Similar to Wang *et al.* [16], Zhao *et al.* [38], Jiang *et al.* [45], Lang and Wu [49], Lang *et al.* [51], Lin *et al.* [52], Xu *et al.* [54], Xu and Lang [55], [56],

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TABLE IV
SAR SHIP CLASSIFICATION ACCURACY OF HOG-SHIPCLSNET

Dataset	Recall (%)	Precision (%)	F1 (%)	LR+	*Accuracy* (%)
OpenSARShip	77.87±1.14	72.42±1.06	75.04±0.68	7.83±0.62	78.15±0.57
FUSAR-Ship	86.62±0.51	86.54±0.50	86.58±0.50	56.59±5.81	86.69±0.47

TABLE V
CONFUSION MATRIX OF HOG-SHIPCLSNET CLASSIFICATION RESULTS ON OPENSARSHIP

Dataset	True \ Predicted	Bulk carrier	Container ship	Tanker	Recall (%)
OpenSARShip	Bulk carrier	231	67	30	70.43
	Container ship	121	646	41	79.95
	Tanker	7	14	125	85.62
	Precision (%)	64.35	88.86	63.78	
	F1 (%)	67.25	84.17	73.10	*Accuracy* = 78.16% [1]
	LR+	5.25	4.68	13.70	

[1] This result is one single experimental realization.

TABLE VI
CONFUSION MATRIX OF HOG-SHIPCLSNET CLASSIFICATION RESULTS ON FUSAR-SHIP

Dataset	True \ Predicted	Bulk carrier	Container ship	Fishing	General cargo	Other cargo	Other	Tanker	Recall (%)
FUSAR-Ship	Bulk carrier	438	23	5	16	8	0	4	88.66
	Container ship	7	495	5	0	6	3	7	94.65
	Fishing	3	1	395	0	44	27	3	83.51
	General cargo	3	0	0	511	0	0	1	98.84
	Other cargo	12	2	89	4	337	57	20	64.68
	Other	0	2	24	0	17	475	2	91.35
	Tanker	5	5	1	8	10	3	489	93.86
	Precision (%)	93.59	93.75	76.11	94.81	79.86	84.07	92.97	
	F1 (%)	91.06	94.20	79.64	96.78	71.47	87.56	93.41	*Accuracy* = 87.98% [1]
	LR+	90.82	87.30	20.84	108.15	23.18	30.93	77.27	

[1] This result is one single experimental realization.

Li *et al.* [57], He *et al.* [68], Huang *et al.* [70], Hou *et al.* [71], and Li *et al.* [106], in this article, we mainly adopt accuracy from (15) to analyze the experimental results in Section V and confirm the effectiveness of four mechanisms in HOG-ShipCLSNet in Section VI.

Furthermore, we provide some other metrics, including Recall, Precision, F1, and Likelihood Ratio Positive (LR+).

Recall is defined by [48]

$$\text{Recall} = \frac{\#\text{TP}}{\#\text{TP} + \#\text{FN}}. \tag{16}$$

Precision is defined by [48]

$$\text{Precision} = \frac{\#\text{TP}}{\#\text{TP} + \#\text{FP}}. \tag{17}$$

$F1$ is defined by [48]

$$F1 = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}. \tag{18}$$

LR+ is defined by [107]

$$\text{LR+} = \frac{\text{TPR}}{\text{FPR}} \tag{19}$$

where TPR denotes the true positive rate that is equal to recall in (16) and FPR denotes the false positive rate that is defined by

$$\text{FPR} = \frac{\#\text{FP}}{\#\text{FP} + \#\text{TN}}. \tag{20}$$

Moreover, the confusion matrix [16] is used to show the classification accuracy of each ship category.

TABLE VII
SAR SHIP CLASSIFICATION ACCURACY OF DIFFERENT POLARIZATIONS ON OPENSARSHIP

Type	Recall (%)	Precision (%)	F1 (%)	LR+	*Accuracy* (%)
VV	77.07±1.65	73.45±1.63	75.19±0.72	8.61±1.01	78.32±0.77
VH	78.30±1.11	72.24±1.29	75.14±0.85	7.86±0.78	77.87±0.85
Total	77.87±1.14	72.42±1.06	75.04±0.68	7.83±0.62	78.15±0.57

V. RESULTS

A. SAR Ship Classification Results

1) Results: Table IV shows the ship classification accuracy of HOG-ShipCLSNet on the OpenSARShip and FUSAR-Ship datasets, where ten optimal results are used to obtain the mean and standard deviation, as in [16]. In Table IV, OpenSARShip comprises VV SAR images and VH SAR images. From Table IV, the classification accuracy on OpenSARShip is 78.15%, lower than 86.69% on FUSAR-Ship, because ships in OpenSARShip have low resolutions with inapparent features, leading to lower accuracies.

2) Confusion Matrix: Tables V and VI show the confusion matrix of one single experimental realization. From Tables VII and VIII, most ships can be correctly classified, that is, the value of a diagonal is greater than others in the same line.

3) Discussion on Polarization: In addition, the reports from OpenSARShip [70] indicate that VV and VH polarizations

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TABLE VIII
SAR SHIP CLASSIFICATION ACCURACY WITH DIFFERENT DATA AUGMENTATION TECHNIQUES ON OPENSARSHIP

Type	Recall (%)	Precision (%)	F1 (%)	LR+	*Accuracy* (%)
--	77.30±1.23	69.87±0.97	73.39±0.48	6.92±0.40	76.16±0.74
Rotation	77.59±1.47	71.32±1.50	74.30±0.88	7.40±0.82	77.43±0.89
Flipping	77.64±1.95	71.71±1.88	74.52±0.80	7.69±1.01	77.08±1.23
Cropping	77.87±1.14	72.42±1.06	75.04±0.68	7.83±0.62	78.15±0.57

have different attitudes and each is more sensitive to different conditions, for example, VH is more sensitive to velocity and volume scattering and VV is more sensitive to double bouncing. Hence, we also study their differentiate results. Table VII shows their ship classification accuracies. From Table VII, HOG-ShipCLSNet offers better accuracy for VV polarization than the VH one (78.32% > 77.87%). This phenomenon is also in line with the reports of [70], because ships generally exhibit higher backscattering values in the co-polarization VV channel [14], which is helpful for classification. However, in the cross-polarization VH channel, the backscattering values are lower [14], potentially reducing the classification performance. To be clear, in this article, the total SAR images (i.e., VV + VH) are both selected to train our model, so we can classify ships of different polarizations.

4) Discussion on Data Augmentation: Finally, we also study the influence of different data augmentation techniques, including random rotation, random flipping, and random cropping, on the classification performance. We take all images of VV and VH on OpenSARShip as an example to present results. Table VIII shows their ship classification accuracy results. From Table VIII, all three data augmentation techniques can improve the classification accuracy, that is, 77.43% > 76.16%, 77.08% > 76.16%, and 78.15% > 76.16%. Moreover, random cropping offers the best performance (78.15% > 77.43% > 77.08%), so it is finally adopted in our experiments.

B. Comparison With Modern CNN-Based Methods

1) Accuracy Analysis: Table IX lists the comparison of the quantitative evaluation indices with modern CNN-based methods. From Table IX, HOG-ShipCLSNet obviously outperforms all other methods. For example, on OpenSARShip, the second best accuracy is 74.98% from Huang *et al.* [117], but this is still lower than our HOG-ShipCLSNet result by ∼4%. Thus, HOG-ShipCLSNet achieves a preferable SAR ship classification.

2) Time Analysis: Table X shows the time comparison with CNN-based methods. From Table X, HOG-ShipCLSNet needs 25.75 s to finish training per epoch, which is lower than most other networks. Moreover, the inference time is 4.33 ms per image, which is almost the same as others.

C. Comparison With Traditional Hand-Crafted Feature-Based Methods

1) Accuracy Analysis: Table XI lists the comparison of the quantitative evaluation indices with the traditional hand-crafted feature-based methods. From Table XI, HOG-ShipCLSNet's accuracy is superior to those of the traditional hand-crafted feature methods.

Furthermore, almost all the traditional methods are inferior to the CNN-based ones; however, when traditional HOG features are integrated into our CNN model, HOG-ShipCLSNet can achieve the state-of-the-art classification accuracy, outperforming both modern CNN-based methods and traditional hand-crafted feature-based methods. Thus, we have shown the value of our work in this article.

2) Time Analysis: From Table XI, the inference time per image of HOG-ShipCLSNet is lower than that for all the traditional hand-crafted feature-based methods. This phenomenon is also in line with common sense because the CNN-based models often have faster inference speed than the traditional ones once training is completed. It should be noted that not all the traditional methods need training, so we only compare the inference time, rather than the training time.

D. Receiver Operator Characteristics Metrics

We discuss receiver operator characteristics (ROC) [119] metrics of different methods. Fig. 8 shows the ROC curves of different candidate models on the OpenSARShip and FUSAR-Ship datasets. To avoid curve-color confusion, we plot all the ROC curves in four subfigures.

We also discuss the area under curve (AUC) [119] for different methods, where AUC denotes the area under each ROC curve. Fig. 9 lists the comparison of their AUC results. From Fig. 9, on OpenSARShip and FUSAR-Ship, the AUC values of HOG-ShipCLSNet are superior, showing the state-of-the-art ship classification performance of HOG-ShipCLSNet.

VI. ABLATION STUDY

A. Ablation Study on Mechanism 1: MS-CLS-Mechanism

We conduct some ablation studies to verify the effectiveness of the MS-CLS-Mechanism. We arrange four groups of experiments. Among them, Experiment 1 is directly used to confirm the effectiveness of the MS-CLS-Mechanism, Experiment 2 is used to study the influence on the classification accuracies of different scales, Experiment 3 is used to study the influence on the classification accuracies of different scale-weighting methods, and Experiment 4 is used to study the influence on the classification accuracies of the multi-output and the single-output types.

1) Experiment 1: Effectiveness of MS-CLS-Mechanism: Table XII shows the results of not using the MS-CLS-Mechanism (i.e., only f_5 is used.) and using it. From

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TABLE IX

COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH MODERN CNN-BASED METHODS

Method	OpenSARShip					FUSAR-Ship				
	Recall (%)	Precision (%)	F1 (%)	LR+	*Accuracy* (%)	Recall (%)	Precision (%)	F1 (%)	LR+	*Accuracy* (%)
LeNet-5 [87]	65.15±1.12	60.54±2.47	62.73±1.52	5.37±1.20	65.74±1.50	68.04±0.12	68.02±0.21	68.03±0.15	13.25±0.28	68.36±0.27
AlexNet [81]	68.51±3.04	65.52±1.23	66.94±1.51	6.71±1.15	70.22±0.68	77.11±0.27	77.37±0.40	77.24±0.30	23.83±1.37	77.03±0.27
VGG-11 [83]	73.21±0.96	68.64±1.49	70.85±1.07	7.29±1.11	73.42±0.75	80.83±0.18	80.90±0.22	80.87±0.16	35.00±1.84	80.76±0.17
VGG-13 [83]	72.59±1.29	67.24±1.75	69.79±0.92	6.43±1.08	73.03±0.86	80.88±0.20	80.96±0.23	80.92±0.21	36.68±2.83	80.79±0.20
VGG-16 [83]	68.61±5.33	64.63±2.52	66.53±3.77	6.89±1.23	70.05±1.35	80.53±0.25	80.67±0.23	80.60±0.23	33.26±1.41	80.45±0.22
VGG-19 [83]	60.03±5.18	59.85±5.27	59.89±4.89	7.20±2.54	66.05±1.52	80.32±0.39	80.66±0.58	80.49±0.41	33.92±2.84	80.23±0.38
GoogLeNet [118]	69.73±2.70	68.80±1.81	69.21±1.19	7.44±1.09	73.80±1.32	83.21±0.56	83.37±0.47	83.29±0.50	44.99±5.46	82.98±0.25
ResNet-18 [90]	73.76±1.61	69.40±1.92	71.49±1.04	7.71±1.86	74.64±0.68	79.21±0.22	79.25±0.25	79.23±0.23	25.93±0.90	79.11±0.21
ResNet-34 [90]	71.43±2.72	68.11±1.73	69.69±1.47	7.07±1.13	73.40±1.09	78.84±0.16	78.85±0.18	78.85±0.16	25.95±0.78	78.76±0.14
ResNet-50 [90]	71.67±1.71	66.79±1.27	69.13±1.04	6.30±0.73	72.82±0.75	81.40±0.23	81.36±0.28	81.38±0.25	33.67±1.35	81.31±0.23
ResNet-101 [90]	70.51±2.33	67.21±2.86	68.79±2.26	7.68±1.88	70.85±0.58	80.56±0.23	80.72±0.36	80.64±0.29	31.83±2.44	80.49±0.23
ResNet-152 [90]	68.88±2.01	63.74±1.70	66.18±0.95	5.75±1.05	69.02±0.70	80.19±0.27	80.19±0.28	80.19±0.27	29.63±1.92	80.10±0.26
ResNext-50-32x4d [91]	70.30±2.14	66.04±2.15	68.06±1.19	6.42±1.62	71.65±0.67	80.64±0.35	80.83±0.37	80.74±0.35	32.54±3.40	80.54±0.36
ResNext-101-32x8d [91]	70.78±2.77	67.05±1.93	68.81±1.21	6.78±1.86	72.61±0.93	80.84±0.31	80.93±0.27	80.89±0.28	32.23±2.54	80.78±0.29
Wide-ResNet-50 [91]	73.45±1.96	67.64±1.95	70.40±1.26	6.67±1.45	73.28±1.17	81.20±0.36	81.29±0.37	81.25±0.36	32.35±2.68	81.15±0.35
Wide-ResNet-101 [91]	72.12±2.77	67.37±1.46	69.62±1.27	6.83±1.61	73.04±0.73	81.02±0.54	81.10±0.51	81.06±0.52	31.43±1.10	80.98±0.53
DenseNet-121 [109]	72.55±3.88	69.56±2.17	70.93±1.60	7.77±1.99	74.65±0.68	82.67±0.37	83.00±0.53	82.83±0.44	38.24±3.87	82.62±0.36
DenseNet-161 [109]	72.54±3.39	67.77±1.46	70.02±1.51	6.82±1.23	73.39±0.79	83.53±0.30	83.57±0.31	83.55±0.30	40.26±4.67	83.45±0.28
DenseNet-169 [109]	71.40±1.80	68.83±1.50	70.07±1.00	7.34±1.05	74.31±0.76	82.34±0.45	82.45±0.46	82.40±0.45	35.21±1.86	82.25±0.13
DenseNet-201 [109]	72.07±3.33	68.29±1.98	70.07±1.73	7.69±2.16	72.96±0.66	82.61±0.33	82.86±0.38	82.73±0.35	37.70±2.56	82.55±0.33
MobileNet-v1 [110]	66.30±2.87	63.49±2.40	64.83±2.03	5.85±1.47	69.91±1.08	77.79±0.56	77.92±0.59	77.86±0.56	24.75±1.77	77.61±0.54
MobileNet-v2 [111]	66.01±4.05	61.80±2.55	63.82±3.15	6.18±0.94	65.83±0.81	74.10±3.02	74.10±3.02	74.10±3.02	19.50±2.60	74.63±0.35
MobileNetv3-Large [112]	65.12±2.53	60.75±1.72	62.84±1.73	5.48±0.94	66.13±0.92	71.00±0.39	70.90±0.42	70.95±0.39	16.15±0.85	70.79±0.37
MobileNetv3-Small [112]	67.23±1.59	61.85±1.69	64.42±1.41	5.52±0.70	66.71±0.87	66.31±0.27	66.12±0.31	66.21±0.26	12.59±0.49	66.02±0.26
SqueezeNet-v1.0 [113]	71.47±1.31	66.73±1.70	69.01±1.28	6.61±1.03	72.15±1.25	78.87±0.40	79.07±0.52	78.97±0.44	29.68±2.05	78.76±0.38
SqueezeNet-v1.1 [113]	67.42±4.67	65.67±1.87	66.45±2.61	7.39±2.44	70.89±1.11	76.35±0.26	76.44±0.29	76.39±0.27	23.26±1.15	76.20±0.21
Inception-v3 [114]	67.13±1.41	63.03±1.63	65.00±1.20	6.14±0.89	65.99±1.08	74.70±0.53	74.84±0.62	74.77±0.56	19.86±1.55	74.61±0.46
Inception-v4 [115]	69.26±3.16	67.43±2.39	68.28±1.97	7.53±1.64	72.44±0.70	80.55±0.40	80.89±0.55	80.72±0.45	32.15±2.92	80.50±0.37
Xception [116]	71.56±3.00	68.60±1.67	70.00±1.29	7.47±1.60	73.74±0.86	77.42±0.39	77.39±0.36	77.41±0.37	22.91±0.95	77.29±0.38
Wang *et al.* [61]	57.72±1.37	58.72±4.76	58.12±2.67	7.42±1.06	69.27±0.27	80.09±0.47	80.32±0.67	80.21±0.56	32.91±2.32	80.02±0.46
Hou *et al.* [71]	69.33±2.00	69.44±2.42	66.76±1.64	6.74±1.76	67.41±1.13	71.96±0.56	72.19±0.59	72.08±0.56	17.79±1.00	71.89±0.54
Huang *et al.* [117]	74.74±1.60	69.56±2.38	72.04±1.60	7.39±1.43	74.98±1.46	83.19±0.41	83.34±0.31	83.27±0.35	42.44±2.52	83.19±0.31
HOG-ShipCLSNet	**77.87±1.14**	**72.42±1.06**	**75.04±0.68**	**7.83±0.62**	**78.15±0.57**	**86.62±0.51**	**86.54±0.50**	**86.58±0.50**	**56.59±5.81**	**86.69±0.47**

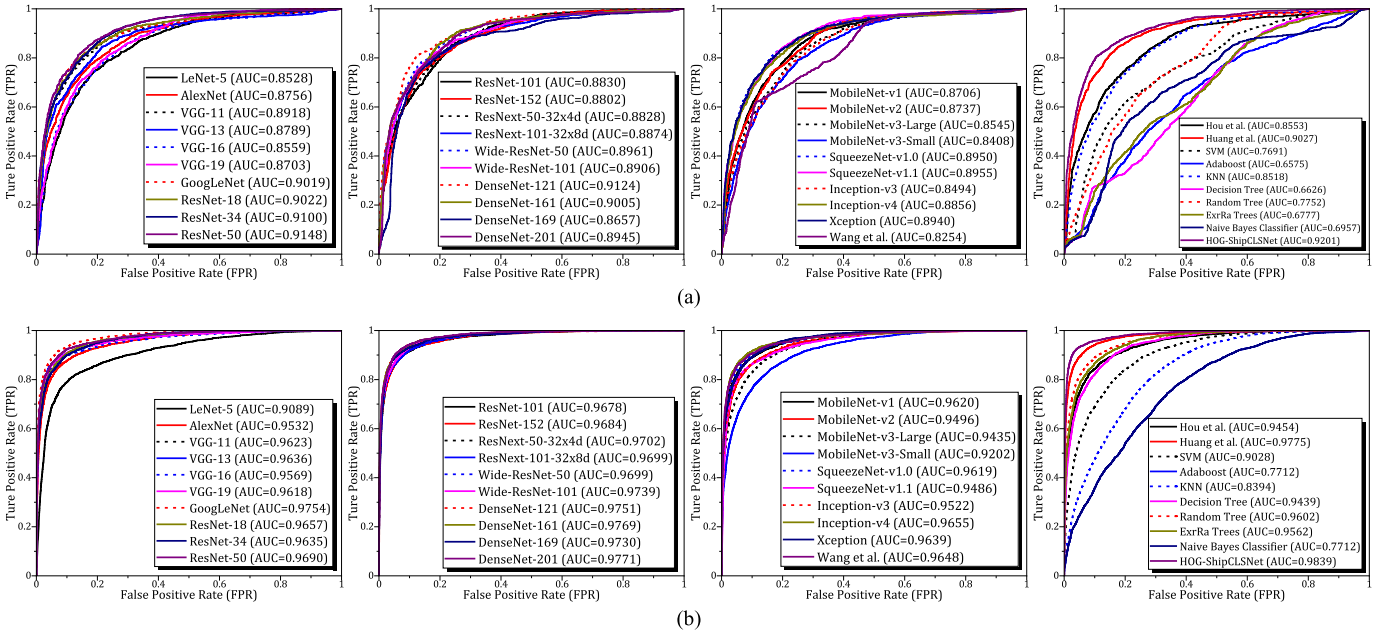


Fig. 8. ROC curves for different methods. (a) ROC curves on OpenSARShip. (b) ROC curves on FUSAR-Ship.

Table XII, when using the MS-CLS-Mechanism, the classification accuracy is improved by ∼5% on OpenSARShip and ∼4% on FUSAR-Ship, showing the MS-CLS-Mechanism's effectiveness.

2) Experiment 2: Different Scales: Table XIII shows the accuracy results on different scales used in HOG-ShipCLSNet. From Table XIII, the accuracy increases along with the number of scales, and when all the scale features are jointly used

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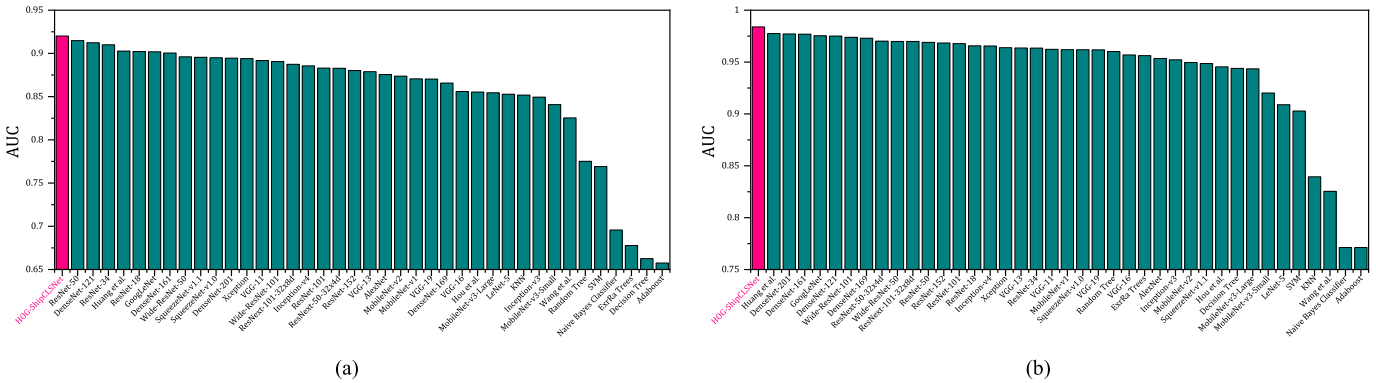


Fig. 9. AUC of different methods. (a) AUC on OpenSARShip. (b) AUC on FUSAR-Ship. Note: Sort AUC values from large to small.

TABLE X
TIME COMPARISON WITH MODERN CNN-BASED METHODS

Method	Training time per epoch (s)	Inference time per image (ms)
LeNet-5 [87]	25.37	4.21
AlexNet [81]	24.65	4.27
VGG-11 [83]	25.93	4.23
VGG-13 [83]	29.41	4.24
VGG-16 [83]	32.13	4.33
VGG-19 [83]	34.51	4.36
GoogLeNet [118]	30.13	4.22
ResNet-18 [90]	25.81	4.25
ResNet-34 [90]	25.89	4.29
ResNet-50 [90]	31.04	4.32
ResNet-101 [90]	44.84	4.36
ResNet-152 [90]	55.78	4.38
ResNext-50-32x4d [91]	34.48	4.28
ResNext-101-32x8d [91]	74.77	4.35
Wide-ResNet-50 [91]	52.92	4.48
Wide-ResNet-101 [91]	85.71	4.61
DenseNet-121 [109]	50.88	4.30
DenseNet-161 [109]	71.29	4.35
DenseNet-169 [109]	66.87	4.36
DenseNet-201 [109]	76.65	4.40
MobileNet-v1 [110]	28.70	4.44
MobileNet-v2 [111]	27.19	4.32
MobileNet-v3-Large [112]	34.46	4.36
MobileNet-v3-Small [112]	29.72	4.32
SqueezeNet-v1.0 [113]	24.67	4.29
SqueezeNet-v1.1 [113]	24.60	4.28
Inception-v3 [114]	40.71	4.34
Inception-v4 [115]	52.97	4.32
Xception [116]	34.29	4.37
Wang *et al.* [61]	29.14	4.33
Hou *et al.* [71]	26.53	4.30
Huang *et al.* [117]	50.48	4.28
HOG-ShipCLSNet	25.75	4.33

for the final classification, the accuracy reaches its highest value. Thus, the multiscale classification concept combining different level features is rather useful for SAR ship classification.

3) Experiment 3: Different Weighting Types of Scales: Table XIV shows the accuracy results of different scale weighting methods. In (1), the average weighting style is adopted in HOG-ShipCLSNet, which can be roughly expressed as

$$I_{\mathrm{MS-CLS}} \leftarrow 0.2 \cdot I''_1 + 0.2 \cdot I''_2 + 0.2 \cdot I''_3 + 0.2 \cdot I''_4 + 0.2 \cdot I''_5. \quad (21)$$

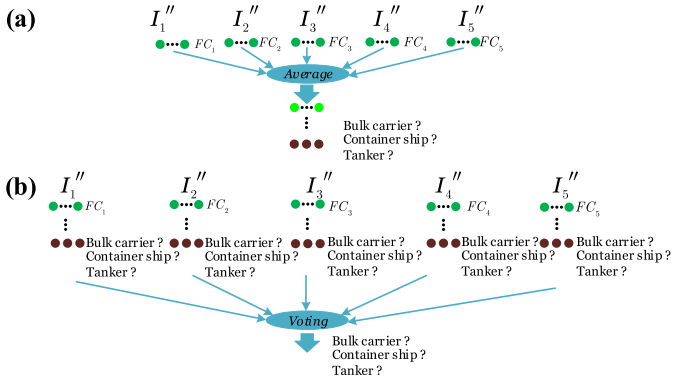


Fig. 10. Output types. (a) Single-output (Ours). (b) Multi-output (the voting mechanism is used).

We also design an adaptive weighting type for comparison, that is,

$$I_{\mathrm{MS-CLS}} \leftarrow w_1 \cdot I''_1 + w_2 \cdot I''_2 + w_3 \cdot I''_3 + w_4 \cdot I''_4 + w_5 \cdot I''_5 \quad (22)$$

where w_1, w_2, w_3, w_4, and w_5 denote the adaptive learning weights of different scales and $w_1 + w_2 + w_3 + w_4 + w_5 = 1$. In our experiments, we add another neuron into the terminal FC layer to adaptively achieve the above five weight values during training; moreover, the five weighting values are inputted into an extra softmax layer to ensure that their sum equals 1.

From Table XIV, unexpectedly, one can find that the average weighting type achieves a slightly better accuracy than the adaptive type. One possible reason is that if the adaptive type is used, the network might pay more attention to search the suitable weight parameter, which may lead to ship feature extraction insufficiency. Another possible reason may be that during training, the adaptive weighting style may pose small training gradients if maintaining the same training epochs, and such a phenomenon might have slower learning, leading to reduced accuracy. More deep-seated reasons can be studied in the future.

4) Experiment 4: Different Output Types: Table XV shows the accuracy results of the multi-output and single-output types. Fig. 10 shows their respective implementation

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TABLE XI
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH TRADITIONAL HAND-CRAFTED FEATURE-BASED METHODS

Method	OpenSARShip					FUSAR-Ship					Inference time
	Recall (%)	Precision (%)	F1 (%)	LR+	*Accuracy* (%)	Recall (%)	Precision (%)	F1 (%)	LR+	*Accuracy* (%)	per image (ms)
SVM [122]	60.23	52.97	56.37	2.95	59.20	60.43	59.81	60.12	9.46	60.04	40.76
Adaboost [123]	56.67	49.19	52.67	2.64	50.08	36.70	34.20	35.41	3.22	36.85	35.62
KNN [124]	65.11	60.20	62.56	4.61	64.51	47.03	48.37	47.69	5.94	46.54	27.77
Decision Tree [125]	48.73	43.49	45.96	1.88	46.02	69.65	69.16	69.40	14.58	69.43	5.50
Random Forest [126]	55.55	48.93	52.03	2.53	53.04	77.33	77.65	77.49	28.82	77.19	9.37
ExrRa Trees [127]	52.58	46.04	49.10	2.20	46.65	75.25	75.98	75.61	25.24	75.12	9.15
Naive Bayes Classifier [128]	48.79	46.92	47.84	2.44	56.24	37.31	34.52	35.86	3.35	36.70	5.72
HOG-ShipCLSNet	**77.87**	**72.42**	**75.04**	**7.83**	**78.15**	**86.62**	**86.54**	**86.58**	**56.59**	**86.69**	**4.33**

TABLE XII
ABLATION STUDY ON MECHANISM 1: MS-CLS-MECHANISM

Dataset	MS-CLS-Mechanism	*Accuracy* (%)
OpenSARShip	×	74.94±0.75
	✓	**78.15±0.57**
FUSAR-Ship	×	81.38±0.76
	✓	**86.69±0.47**

TABLE XIII
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH DIFFERENT NUMBER SCALES IN THE MS-CLS-MECHANISM

Dataset	f_1	f_2	f_3	f_4	f_5	*Accuracy* (%)
OpenSARShip	×	×	×	×	✓	74.94±0.75 [1]
	×	×	×	✓	✓	75.76±0.98
	×	×	✓	✓	✓	76.79±0.46
	×	✓	✓	✓	✓	77.08±0.95
	✓	✓	✓	✓	✓	**78.15±0.57** [2]
FUSAR-Ship	×	×	×	×	✓	81.38±0.76 [1]
	×	×	×	✓	✓	84.72±0.60
	×	×	✓	✓	✓	84.96±0.38
	×	✓	✓	✓	✓	85.75±0.34
	✓	✓	✓	✓	✓	**86.69±0.47** [2]

[1] Only using f_5 is equivalent to completely omitting MS-CLS-Mechanism, i.e., "×" in Table XII.
[2] Using f_1, f_2, f_3, f_4 and f_5 is equivalent to completely adopting MS-CLS-Mechanism, i.e., "✓" in Table XII.

TABLE XIV
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH DIFFERENT WEIGHTING TYPES

Dataset	Weighting Type	*Accuracy* (%)
OpenSARShip	Adaptive	77.33±0.59
	Average (Ours)	**78.15±0.57**
FUSAR-Ship	Adaptive	85.33±0.30
	Average (Ours)	**86.69±0.47**

processes. As mentioned in Section II-A, the single-output type is selected in our HOG-ShipCLSNet.

From Table XV, the single-output type can achieve superior accuracy to the multi-output one, because the latter has the risk of training oscillation; moreover, it may also lead to decision failure, for example, for a three-classification task (i.e., classes 0, 1, and 2). If the outputs from five scales are class 0, class 0, class 1, class 1, and class 2, then the network will fall into training oscillation due to decision failure if using a voting mechanism (i.e., confusion between classes 0 and 1). Therefore, our single-output type is more suitable for classification tasks, which differ from the detection tasks in FPN.

TABLE XV
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH DIFFERENT OUTPUT TYPES

Dataset	Output Type	*Accuracy* (%)
OpenSARShip	Multi Output	76.80±0.51
	Single Output (Ours)	**78.15±0.57**
FUSAR-Ship	Multi Output	85.70±0.45
	Single Output (Ours)	**86.69±0.47**

TABLE XVI
ABLATION STUDY ON MECHANISM 2: GS-ATT-MECHANISM

Dataset	GS-ATT-Mechanism	*Accuracy* (%)
OpenSARShip	×	76.89±0.45
	✓	**78.15±0.57**
FUSAR-Ship	×	84.83±0.55
	✓	**86.69±0.47**

B. Ablation Study on Mechanism 2: GS-ATT-Mechanism

We conduct some ablation studies to verify the effectiveness of the GS-ATT-Mechanism. We arrange two groups of experiments. Among them, Experiment 1 is used directly to confirm the effectiveness of GS-ATT-Mechanism and Experiment 2 is used to study the influence on the classification accuracies of the different attention types, including the commonly used types: the convolutional block attention module (CBAM) [120] and the Squeeze-and-Excitation (SE) attention mechanism [121].

1) Experiment 1: Effectiveness of GS-ATT-Mechanism: Table XVI shows the accuracy results using the GS-ATT-Mechanism and not using it. From Table XVI, when using the GS-ATT-Mechanism, the classification accuracy is improved by ∼2% on both OpenSARShip and FUSAR-Ship, because ship features with global self-attention will be more discriminative. Thus, the above fully shows the GS-ATT-Mechanism's effectiveness.

2) Experiment 2: Different Attention Types: Table XVII shows the accuracy results of different attention types. From

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TABLE XVII
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH DIFFERENT ATTENTION TYPES

Dataset	Attention Type	*Accuracy* (%)
OpenSARShip	CBAM [120]	77.01±0.94
	SE [121]	76.61±0.40
	GS-ATT-Mechanism (Ours)	**78.15±0.57**
FUSAR-Ship	CBAM [120]	86.07±0.45
	SE [121]	85.31±0.36
	GS-ATT-Mechanism (Ours)	**86.69±0.47**

TABLE XVIII
ABLATION STUDY ON MECHANISM 3: FC-BAL-MECHANISM

Dataset	FC-BAL-Mechanism	*Accuracy* (%)
OpenSARShip	×	74.87±0.81
	✓	**78.15±0.57**
FUSAR-Ship	×	71.55±0.66
	✓	**86.69±0.47**

Table XVII, our GS-ATT-Mechanism achieves a better accuracy than CBAM and SE, coming from more discriminative ship features, by which more useful ship information is well-maintained well, as mentioned above.

C. Ablation Study on Mechanism 3: FC-BAL-Mechanism

We conduct some ablation studies to verify the effectiveness of the FC-BAL-Mechanism. Here, we arrange two groups of experiments. Experiment 1 is used directly to confirm the effectiveness of the FC-BAL-Mechanism and Experiment 2 is used to determine the neuron numbers N_{FC} in our FC-BAL-Mechanism.

1) Experiment 1: The effectiveness of FC-BAL-Mechanism. Table XVIII shows the accuracy results using the FC-BAL-Mechanism (i.e., adding another five FC layers) and not using it (i.e., not adding another five FC layers). From Table XVIII, when using the FC-BAL-Mechanism, the accuracy is greatly improved, that is, ∼4% on OpenSARShip and ∼5% on FUSAR-Ship, fully showing the FC-BAL-Mechanism's effectiveness.

2) Experiment 2: Different Neuron Numbers NFC: Table XIX shows the accuracy results of different neuron numbers N_{FC}. From Table XIX, with the increase in N_{FC}, the accuracy first rises and then falls or saturates.

On the OpenSARShip dataset, when $N_{\mathrm{FC}} = 1024$, the accuracy reaches the maximum, and on the FUSAR-Ship dataset, when $N_{\mathrm{FC}} = 1152$, the accuracy is maximal. Therefore, in actual experiments, one can adjust it and then select an optimal value of N_{FC} to achieve the best classification performance, where we suggest $N_{\mathrm{FC}} \approx 1000$.

D. Ablation Study on Mechanism 4: HOG-FF-Mechanism

We conduct some ablation studies to verify the effectiveness of the HOG-FF-Mechanism. Here, we arrange four groups of experiments. Experiment 1 is used directly to confirm the effectiveness of the HOG-FF-Mechanism, Experiment 2 is used to confirm the necessity of PCA, Experiment 3 is used to determine the best PCA dimension, and Experiment 4 is used to study different traditional hand-crafted feature fusion methods, where canny edge features [129], local binary patterns (LBPs) features [130], corner Harris features [131], and texture features [132] are selected for comparison.

TABLE XIX
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH DIFFERENT NUMBERS OF NEURONS IN THE FC-BAL-MECHANISM

Dataset	N_{FC}	*Accuracy* (%)
OpenSARShip	128	74.70±1.29
	256	75.76±1.17
	384	77.22±0.73
	512	75.61±0.83
	640	77.16±1.07
	768	77.17±0.72
	896	77.46±0.47
	1024	**78.15±0.57**
	1152	76.29±0.62
	1280	76.76±0.76
	1408	75.36±1.58
	1536	76.79±0.67
	1664	76.44±0.40
	1792	76.69±0.87
	1920	75.98±0.84
	2048	74.34±0.96
FUSAR-Ship	128	75.78±0.62
	256	81.42±0.35
	384	82.41±0.39
	512	82.93±0.63
	640	84.10±0.32
	768	84.46±0.38
	896	84.37±0.52
	1024	84.83±0.32
	1152	**86.69±0.47**
	1280	85.04±0.35
	1408	85.00±0.42
	1536	85.22±0.32
	1664	85.34±0.44
	1792	85.40±0.57
	1920	85.25±0.34
	2048	85.16±0.33

TABLE XX
ABLATION STUDY ON MECHANISM 4: HOG-FF-MECHANISM

Dataset	HOG-FF-Mechanism	*Accuracy* (%)
OpenSARShip	×	74.95±0.79
	✓	**78.15±0.57**
FUSAR-Ship	×	75.34±0.16
	✓	**86.69±0.47**

1) Experiment 1: Effectiveness of HOG-FF-Mechanism: Table XX shows the accuracy results with and without HOG-FF-Mechanism. From Table XX, when using the HOG-FF-Mechanism, the accuracy achieves huge improvements, that is, ∼4% on OpenSARShip and ∼10% on FUSAR-Ship, demonstrating its effectiveness.

Therefore, traditional mature hand-crafted features can also play an important role in the CNN-based models, showing that the practice that excessive reliance on abstract features by other existing CNN-based SAR ship classifiers may be controversial. In fact, some scholars from the SAR ship detection community have recently realized this problem, for

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TABLE XXI
ABLATION STUDY ON PCA IN HOG-SHIPCLSNET

Dataset	PCA	*Accuracy* (%)
OpenSARShip	×	76.43±0.60
	✓	**78.15±0.57**
FUSAR-Ship	×	85.95±0.23
	✓	**86.69±0.47**

TABLE XXII
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH DIFFERENT DIMENSIONS OF PCA

Dataset	k	*Accuracy* (%)
OpenSARShip	32	76.06±0.67
	64	76.13±0.61
	96	76.75±0.95
	128	77.37±1.10
	160	76.13±0.97
	192	77.15±0.77
	224	76.18±0.67
	256	**78.15±0.57**
	288	76.78±1.00
	320	77.64±0.58
	352	77.76±0.53
	384	76.98±1.30
	416	75.09±1.18
	448	76.87±0.63
	480	75.26±1.04
	512	75.65±0.58
FUSAR-Ship	32	85.93±0.18
	64	85.99±0.24
	96	85.96±0.23
	128	85.58±0.26
	160	**86.69±0.47**
	192	85.43±0.25
	224	86.04±0.37
	256	85.74±0.26
	288	85.94±0.26
	320	86.04±0.32
	352	86.14±0.22
	384	85.10±0.51
	416	86.25±0.14
	448	85.95±0.14
	480	85.93±0.26
	512	86.27±0.56

example, Ai *et al.* [25] and Huang *et al.* [66], [67], which also shows the rationality of our idea of HOG feature fusion.

2) Experiment 2: Necessity of PCA: Table XXI shows the accuracy results with and without PCA. From XXI, PCA dimension reduction can effectively improve the classification accuracy, that is, ∼2% on OpenSARShip and ∼1% on FUSAR-Ship, because the original HOG features have some redundancy due to their huge dimensionality (32 448-D). If there are too many features that are not widely representative to be joined, the network will have a certain risk of overfitting. In short, the above shows the necessity of PCA.

3) Experiment 3: Different PCA Dimension k: Table XXII shows the results for different PCA dimensions k. From Table XXII, as k increases, the accuracy presents a state of first rising and then falling, or saturation. For OpenSARShip, the accuracy is maximal when $k = 256$; for FUSAR-Ship, this maximum comes when $k = 160$. Thus, one can adjust an optimal value of k to achieve the best classification performance for $k \approx 200$. In addition, traditional features might contain the less representative ship information than the CNN-based abstract features, so its feature dimension is smaller than the CNN-based abstract feature dimension. It should be noted that the training of our HOG-ShipCLSNet is not sensitive to choices of k and different k values can all ensure training stability.

TABLE XXIII
COMPARISON OF QUANTITATIVE EVALUATION INDICES WITH DIFFERENT TRADITIONAL HAND-CRAFTED FEATURES

Dataset	Feature	*Accuracy* (%)
OpenSARShip	Canny edge [129]	77.57±0.86
	LBP [130]	58.62±0.97
	Corner Harris [131]	77.33±0.43
	Texture [132]	63.10±1.48
	HOG (Ours)	**78.15±0.57**
FUSAR-Ship	Canny edge [129]	85.24±0.49
	LBP [130]	70.71±0.36
	Corner Harris [131]	85.20±0.14
	Texture [132]	79.31±0.32
	HOG (Ours)	**86.69±0.47**

4) Experiment 4: Different Traditional Hand-Crafted Feature Fusions: Table XXIII shows the accuracy results of different traditional hand-crafted feature fusion methods. From Table XXIII, our used HOG feature fusion outperforms the other feature fusion, which shows that the HOG features are superior, as is in line with the work of Lin *et al.* [52]. In addition, one can also find that canny edge features and corner Harris features can improve the CNN-based models' performance; however, LBP features and texture features pose some unexpected adverse impacts on classification performance. One possible reason may be that the two features might have a rejection reaction of CNN abstract features. More deep-seated reasons need to be further studied and discussed in the future. Therefore, selecting the suitable traditional features for CNN feature fusion remains a meaningful question worth studying in the future.

VII. CONCLUSION

Aiming at the circumstance by which most existing CNN-based SAR ship classifiers rely excessively on abstract features while uncritically abandoning traditional hand-crafted ones to their detriment, a novel HOG-ShipCLSNet is proposed for preferable SAR ship classification. In HOG-ShipCLSNet, four novel mechanisms (MS-CLS-Mechanism, GS-ATT-Mechanism, FC-BAL-Mechanism, and HOG-FF-Mechanism) are proposed to guarantee HOG-ShipCLSNet's state-of-the art classification performance. In this article, for each mechanism, we all conduct sufficient ablation studies to: 1) verify its effectiveness and 2) confirm its feasibility including the determination of experimental parameters and various essential discussions. Finally, on the two open ship datasets (OpenSARShip and FUSAR-Ship), HOG-ShipCLSNet is observably superior to both modern CNN-based methods and traditional hand-crafted feature ones. Thus, HOG-ShipCLSNet

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can indeed reliably and truly achieve preferable SAR ship classification.

Our future work is as follows.

1) We will study how to select the most suitable traditional features for CNN feature fusion in the future.
2) We will rethink and analyze the deep-seated internal mechanisms of traditional feature fusion in the future.
3) We will study hybrid feature fusion forms in the future, potentially improving the accuracy.
4) We will analyze the differences in network training between the average weighting style and the adaptive weighting one in the MS-CLS-Mechanism for possible future improvements.

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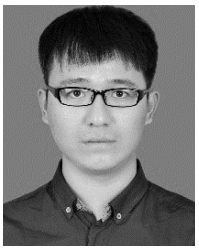
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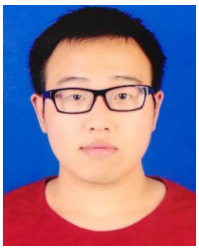
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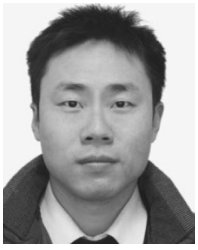
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